

FT-ITC Analysis User manual

Processing, fitting, and interpreting isothermal titration calorimetry experiments.

VERSION 1.5.0 • FOR MACOS, WINDOWS, AND LINUX

Contents

01 FT-ITC Analysis user manual

Start here for a tour of FT-ITC Analysis and the conventions used throughout this manual.

02 Quick start

Open your own ITC dataset, process it, fit a one-set-of-sites model, review the result, save the project, and export a figure.

03 Installation, files, and projects

Install FT-ITC Analysis, open supported data, save portable projects, and use autosave or recovery safely.

04 Workspace

The application window, Experiment Data capabilities, and Analysis Result capabilities.

05 Processing

Baseline models, integration regions, injection uncertainty, and processing propagation and locking.

06 Single-experiment fitting

Fit one experiment, configure model and uncertainty options, control injection inclusion, and interpret fit diagnostics.

07 Multiple-experiment fitting

Multi-dataset fitting across processed Active experiments, including constraints, shared options, diagnostics, and combined results.

08 Results and advanced analyses

Analysis Result views, validity, uncertainty presentation, evaluation temperature, and conditional advanced analyses.

09 Figures and export

Configure final and supporting figures, print active graphs, and export data, results, figures, and complete analysis reports.

10 Tools

Design simulated titrations, subtract buffer controls, and merge standard or back-mixed tandem experiments.

11 Settings and defaults

Configure shared defaults for display, processing, fitting, and export.

12 Glossary and equations

Terminology, keyboard conventions, calculation symbols, equations, and references.

CHAPTER 01

FT-ITC Analysis user manual

Start here for a tour of FT-ITC Analysis and the conventions used throughout this manual.

Last verified: 2026-09-23

In isothermal titration calorimetry (ITC), an instrument measures heat as one solution is added to another. FT-ITC Analysis turns those measurements into heat values for individual injections, fits a proposed explanation of the interaction, and helps you inspect and present the result.

This manual is for ITC practitioners who are new to FT-ITC Analysis. It explains how to use the application and provides guidance on scientific interpretation; no programming knowledge is required.

These instructions apply on macOS, Windows, and Linux. The manual treats the supported desktop applications as one product. A **Platform note** appears only when an operating-system or interface difference changes how you complete a task.

VERIFIED:

This edition reflects FT-ITC Analysis 1.5.0 and was verified on the date shown for each page.

What the application does

The workflow has two main questions: **How much heat did each injection produce?** and **What interaction could explain those heats?** Processing answers the first question from a raw power trace (a **thermogram**). Fitting compares the injection heats with the predictions of a chosen model. A close fit is useful evidence, but it does not prove that the model describes the chemistry.

FT-ITC Analysis supports the complete workflow from compatible instrument data to a saved analysis project and publication figures:

1. Import a raw thermogram, or integrated heat files from multiple different sources.
2. Review and edit experiment details, concentrations, comments, and attributes.
3. Process thermograms using baseline models, spline editing, and manual or automatic adjustment of integration regions.
4. Fit one or more experiments with a supported model.
5. Review fitted parameters, uncertainties, residuals, statistics, and perform further analysis.
6. Save portable projects in the `.ftxtc` format.
7. Export figures, numerical data, and reports.

Raw input files are read, and never modified. Desktop analysis is local: experiment data are not uploaded during ordinary analysis. An optional launch-time online check retrieves version and citation information. Optional automated interpretation sends the selected report evidence and supplied context to the hosted interpretation service at app.ft-itc.org, which uses OpenAI; usage metadata are retained without automatic expiry. The separate browser-based Project Viewer at the same host uploads a project for viewing and parsing; opening a project there does not submit it for automated interpretation. Uploaded data is not retained on the server. See Privacy and online checks and Analysis Report.

Product tour

The data list contains loaded experiments and completed Analysis Results. Selecting an experiment exposes four shared task views:

- **Overview** summarizes the experiment and provides access to its details.
- **Process Data** builds and refines baselines, adjusts injection regions, and converts a raw thermogram into integrated heats with estimated uncertainties.
- **Analyze Data** fits a single experiment or multiple experiments.
- **Final Figure** presents the thermogram, heats, fitted curve, residuals, and annotations.

Selecting an Analysis Result opens its result workspace, with a parameter summary, member fits, parameter correlation, uncertainty display, and any compatible advanced analyses. The menus provide project operations, experiment management, export commands, preferences, additional tools, citation information, and support links. The FT-ITC Analysis website, latest release, and software DOI provide project, installation, and citation context.

Manual conventions

The shortest route to a first result is Quick start, which uses your own compatible data. The relevant task chapter provides setting details and diagnostic guidance when a workflow needs closer inspection.

Manual directory

- Quick start
- Installation, files, and projects
- Workspace
- Processing
- Single-experiment fitting
- Multiple-experiment fitting
- Results and advanced analyses
- Figures and export
- Tools
- Settings and defaults
- Glossary and equations

The manual uses these callouts:

NOTE:

A detail that affects how the application behaves.

CAUTION:

A condition that can produce a failed task, invalid result, or misleading output.

INTERPRETATION:

Scientific context for judging output; it is not an automated conclusion made by the application.

RECOMMENDATION:

A defensible working practice rather than a software requirement.

PLATFORM NOTE:

A genuine operating-system or interface difference that changes how a task is completed.

CALCULATION:

A compact relationship used by FT-ITC Analysis or needed to interpret its output.

Interface labels appear in **bold**. A path such as **File > Save As...** means choose the menu and then the command. The names of views and controls are used instead of position-dependent phrases, so the instructions remain usable when the window size or platform changes.

Help and support

Help and Guide opens an in-app manual, and **Technical Details** opens the in-app processing and scientific notes. **Citation** presents the current paper and versioned software citations and supports copying or exporting BibTeX. **Contact Support...** helps you contact support by email and prepares a diagnostic report containing the application version, operating system, recent activity, and full application log. **Copy Support Report** places that report on the clipboard. Official links include the FT-ITC Analysis website, latest release, project viewer, and software DOI 10.5281/zenodo.14832177.

CHAPTER 02

Quick start

Open your own ITC dataset, process it, fit a one-set-of-sites model, review the result, save the project, and export a figure.

Last verified: 2026-09-23

This walkthrough takes a compatible file from measured heats to a fitted result. A **One-Set-Of-Sites** model is a starting example when the binding sites can reasonably be treated alike; choose another model if the chemistry calls for it. The steps show how to use the application, not how to decide which scientific model is true.



The overall analysis workflow. Integrated-heat imports enter after thermogram processing.

Before you begin

Have one of these inputs available:

- a raw thermogram from a MicroCal-style file (`.itc`), native TA Instruments NanoITC file (`.nita`), NanoAnalyze export (`.ta`), or PEAQ-ITC project (`.apj`).
- an Origin project file (`.opj`). When a recognized ITC worksheet contains the original time/power trace, FT-ITC Analysis restores it for processing; otherwise, it uses the worksheet heat values as integrated input.
- injection-level integrated heats (`.dat` , `.aff` , or `.dh`).
- an FT-ITC project (`.ftxtc`).

Supported formats and project behavior are described in Installation, files, and projects.

NOTE:

Integrated-heat files and Origin projects without an original time/power trace skip baseline correction and peak integration; begin with experiment details and fitting.

1. Open your data

Launch FT-ITC Analysis and choose **Open File...** on the welcome screen, choose **File > Open...**, or drag compatible files into the application window. Select your file and confirm the open operation.

The experiment appears in the data list. Select it and open **Overview** to orient yourself in the imported experiment. You can drag data or result rows to change their order. This order is used throughout the application and is retained when the project is saved as `.ftxtc`.

2. Edit experiment details

Open **Details...** for the selected experiment. Concentration entries and the experiment date/time can be changed here when needed. Comments and attributes relevant to later analysis can also be added or edited. A changed date is marked as user-modified in the experiment overview.

CAUTION:

Apply corrections only when you have an independent experimental basis. Concentration entries influence the calculated concentration ratio and fitted parameters.

3. Process a raw thermogram

If the import contains a thermogram, open **Process Data**.

1. The baseline estimates the signal the instrument would show between injection peaks. If that background drifts smoothly, try **Polynomial**. For a more irregular background, try **Spline**; for changes that differ across parts of the run, try **Segmented**. Keep the default integration settings initially.
2. Inspect whether the baseline represents the signal between injections rather than the peaks.
3. Inspect the start and end of every integration region. A region should include the injection response without extending unnecessarily into baseline noise.

These baseline alternatives are explained in Processing.

RECOMMENDATION:

Zoom to a single peak by double-clicking it, copy the integration region to the next peak using **Space**.

INTERPRETATION:

A visually smooth baseline is generally desired. As a rough rule of thumb, aim for at least 20% of the data points to be baseline data; this is guidance, not a hard threshold. Apparent jumps in the baseline may indicate insufficient equilibration time between injections. Major spikes may require advanced baseline editing; the **Spline** baseline type provides more flexibility.

4. Fit a model

Open **Analyze Data**, choose **Single experiment**, and select **One-Set-Of-Sites** or a more appropriate model, depending on your system.

1. Review the initial parameter values: affinity describes how strongly the substances bind, enthalpy describes the heat change associated with binding, and stoichiometry describes the binding ratio in the chosen model. Use physically plausible starting values.
2. Choose an optimizer. **Levenberg-Marquardt** is efficient near a suitable solution; **Nelder-Mead** provides a derivative-free alternative when convergence is difficult.
3. Optionally enable **Weight by injection error** when the integration uncertainties are meaningful for the dataset.
4. Choose an uncertainty method. **None** skips parameter uncertainty; **Bootstrap residuals** refits simulated variations of the data; **Leave-one-out** checks how much the fit changes when an observation is omitted; **Profile likelihood** tests how far a parameter can move while the fit remains acceptable.
5. Choose **Run Fit**.

Create analysis result determines whether a usable fit also creates a separate Analysis Result. Enable it to continue through the result workspace in the next step; otherwise, the fitted solution remains attached to the Experiment Data.

RECOMMENDATION:

More complex models such as the two-sets-of-sites model may require considerable trial and error with starting parameters in order to obtain a good solution.

If the fit fails or reaches a limit, the possible causes are described under Fit availability and non-convergence.

5. Review the result

Inspect the fitted curve together with the residuals (observed heat minus predicted heat). Random-looking small residuals are more reassuring than a repeated pattern the model misses. When **Create analysis result** is enabled, select the resulting **Analysis Result** and check:

- the included experiment and injections;
- fitted parameter values and units;
- convergence and fit loss;
- whether weighting and uncertainty estimation match your intention;
- parameter uncertainty or confidence intervals, when calculated;
- parameter correlations.

When **Create analysis result** is disabled, the fitted solution remains available in **Analyze Data** rather than as a stored Analysis Result.

INTERPRETATION:

Parameter precision does not establish model adequacy. Look for structured residuals, sensitivity to initial values, values at limits, excessive parameter correlation, or disagreement with known stoichiometry and experimental conditions.

6. Save a portable project

Choose **File > Save As...** and save the project as `.ftxtc`. The project contains imported data, experiment details, processing state, fits, results, and available advanced-analysis output. Keep the original instrument files as source records even though the project is portable.

If autosave is enabled, it supplements normal saving; it is not a replacement for a named project file.

7. Export a figure

Return to the experiment and open **Final Figure**. Choose the elements you need, such as the thermogram, integrated heats, fitted curve, residuals, confidence band, or parameter information. Review axis ranges and labels, then use the figure export or print command provided by the view.

Use **Analysis Result Exporter...** when you need numerical result tables rather than a graphic. For injection-level heats, choose **File > Export Integrated Peaks....** The same output is available through **File > Export Data...** by choosing **Integrated Peaks**.

NOTE:

An Analysis Result can export a figure using its saved fit for each member experiment. Use **Export Associated Final Figures...** on the result to make those figures without first loading the saved solutions into the Experiment Data.

The figure and table workflows continue in Figures and export.

CHAPTER 03

Installation, files, and projects

Install FT-ITC Analysis, open supported data, save portable projects, and use autosave or recovery safely.

Last verified: 2026-09-23

Install the application

Use the latest FT-ITC Analysis release for current packages and release notes. Verify that the file came from this official release page before accepting an operating-system security prompt. Package options may change between releases. For installation help, contact support@ft-itc.org.

macOS

Open the DMG, drag **FT-ITC Analysis** to **Applications**, and eject the disk image. If macOS blocks a verified download, review it in **System Settings > Privacy & Security** and use **Open Anyway** only after confirming its source.

Windows

Run the supplied Windows x64 `.exe` installer and follow the setup prompts. The installer registers `.ftxtc` project associations. If Windows displays an **Unknown publisher** or Microsoft Defender SmartScreen warning, continue only after confirming that the installer came from the official FT-ITC release page.

Linux

Install the supplied `.deb` package matching the system architecture (AMD64 or ARM64, when available) on a compatible Debian-based distribution. Package trust and dependency behavior depend on the channel through which the package was obtained. Review the current release notes for known platform limitations.

PLATFORM NOTE:

Package installation and file-association prompts are controlled by the operating system. The analysis workflow is otherwise shared; the manual notes the few interface labels that differ between desktop editions.

Supported input formats

Choose the input that matches the stage of your work. A **raw thermogram** contains the instrument's power trace and still needs processing. An **integrated-heats** file already contains a heat value for each injection, so it goes directly to fitting. A **project** can reopen earlier FT-ITC work.

EXTENSION	SOURCE	IMPORTED CONTENT
<code>.itc</code>	MicroCal-style raw data	Raw thermogram and injections
<code>.nita</code>	TA Instruments NanoITC native data	Raw thermogram and injections
<code>.ta</code>	TA Instruments/NanoAnalyze export	Raw thermogram and injections
<code>.apj</code>	PEAQ-ITC project	Raw thermogram and injections from the first experiment
<code>.opj</code>	Origin project file	Raw thermogram or integrated heats, according to the recognized worksheet
<code>.dat</code>	Integrated-heats table	Integrated heats and injections
<code>.aff</code>	Integrated-heats table	Integrated heats and injections
<code>.dh</code>	Fixed-layout integrated-heats file	Integrated heats, injections, and experiment metadata
<code>.ftxtc</code>	Current FT-ITC project	Stored project state
<code>.ftitc</code>	Legacy FT-ITC project	Readable for migration; not written by current versions

Raw thermogram imports (`.itc` , `.nita` , `.ta` , and `.apj`) open as unprocessed Experiment Data. Process and fit them in FT-ITC Analysis. A PEAQ-ITC `.apj` import uses only the first experiment; its existing processing and fits are not imported.

For an Origin `.opj` file, FT-ITC Analysis uses the first recognized ITC worksheet. It imports the raw thermogram when available, or integrated heats otherwise. Origin processing and fits are not imported. The newer `.opju` format is unsupported.

If a MicroCal `.itc` file contains concatenated runs, an import prompt offers the standard **Use MicroCal Concat** calculation or **Use Back-Mixing Compensation**. Use the latter when you have the required mixing information. See Experiment Merger for details.

Integrated-heat files (`.dat` , `.aff` , and `.dh`) skip thermogram processing. During import, select the heat unit used by the file and supply any requested concentration information. For `.dat` and `.aff` , **Recalculate concentrations and ratios** replaces the file's concentration progression with one calculated from the injections. Choose **Use selected action for remaining files** only if those files use the same heat units and import choices. These formats contain no thermogram, so they cannot restore a baseline or processing-derived injection uncertainties.

The first injection is imported but excluded from fitting by default because its heat is often unreliable. You can include it manually; its volume still contributes to later concentration calculations.

Open files

Choose **File > Open...**, use the welcome-screen action, or drag files into the application. Multiple supported files can be opened together. Files opened into a populated document are added to its existing Data / Results list; this includes `.ftxtc` projects. Clear the current document first if you want to open a project by itself.

CAUTION:

After an `.ftxtc` project is added to an existing document, that opened project becomes the document's current save destination. Use **Save As...** before saving if you do not intend to replace it with the combined document.

CAUTION:

Appending can create similarly named experiments or results. Confirm the data list and details before fitting or exporting.

Save projects

Choose **File > Save** to update the current named project, or **File > Save As...** to choose a new name or location. Use the current `.ftxtc` format for ongoing work.

An `.ftxtc` project preserves the data and metadata needed to continue analysis, including thermograms where imported, concentrations and uncertainties, attributes and comments, injection inclusion, processing state, fit solutions, Analysis Results, and completed derived analyses. The package is portable and does not depend on the original raw-file path for ordinary reopening.

Save Selected... writes selected project content when you need a smaller handoff. Confirm the selection before saving and reopen the result if the subset is critical. Saving the selected experiments saves only the experiment with any solution. Saving the selected Analysis Results saves the result along with the involved experiments.

RECOMMENDATION:

Save a processed version of the project before fitting if you want a reusable starting point. After fitting, save the project again—under a new name if you want to preserve the processed-only version—to retain the fitted solutions and Analysis Results.

Autosave and recovery

Autosave behavior is configured in **Preferences...** When recovery data is available after an interrupted session, the application offers a recovery path. Open the recovered document, inspect the data list and recent changes, then save it as a named `.ftxtc` project.

Recovery mode is designed to salvage valid project components when possible. If an experiment is unavailable, its saved fit is omitted; Analysis Results that depend on that fit are also omitted rather than being restored with a scientifically different set of members. Unaffected experiments, fits, and results can still be recovered. When recovery was required, the application displays a warning and records the individual recovery issues in the application log. A recovered project can be detached from its former save location and marked as changed. Use **Save As...** rather than assuming the damaged or interrupted file was repaired in place.

CAUTION:

Recovery cannot guarantee that every optional result or saved analysis survived. Confirm experiment counts, processing, fits, and result validity before continuing.

Remove and clear content

Removing an experiment or Analysis Result changes only the open document; it does not delete the original raw file. **Remove All Data/Results** clears the current document after confirmation. **Clear Processing/Results** removes all Analysis Results from the open document.

Saving after removal makes the removal part of the saved project. Use **Save As...** first if you want to preserve the original project version.

Privacy and online checks

Processing, fitting, saving, recovery, export, and printing run locally. Optional automated interpretation in **Analysis Report** sends the selected report evidence, question, and context to an online service when you request generation. See Analysis Report for details.

If **Check for updates and online resources on launch** is enabled, the application checks for release and citation updates; it does not upload experiment data. Disable this setting to prevent launch-time checks; it does not control automated interpretation requests. A failed or disabled check does not prevent local processing, fitting, or saving.

Update safely

Before installing a new application version, save important projects and retain the current installer when reproducibility policy requires it. After updating, open a copy of a representative project, confirm processing and results, and save only when you intend the project to be written by the new version.

Online data flow and retention

Automated interpretation: Choosing **Generate** sends the selected report evidence and your question or context to FT-ITC's online interpretation service at app.ft-itc.org. The service uses OpenAI to generate the interpretation. Sent evidence can include experiment and result names, comments, conditions, concentrations, saved fits, uncertainty summaries, diagnostics, and injection data. Supporting experiments are included when selected. Compressed thermograms are off when the dialog opens and are sent only if you opt in. **Save package** creates a local copy for review without sending it. Opening Preferences or the generation dialog can check service availability and account access without sending scientific evidence. A supplied access code is sent to verify access.

Project Viewer: Opening a project in the browser-based Project Viewer uploads the complete `.ftxtc` file, including its embedded data and comments, to app.ft-itc.org for display. This does not submit the project for automated interpretation or send it to OpenAI. The viewer has a 50 MB upload limit. Closing the viewer does not retract an upload or guarantee that temporary processing copies or infrastructure records have been deleted; the full service has no verified end-to-end deletion schedule.

Retention: FT-ITC's interpretation service retains usage metadata that can be linked to requests or accounts, without automatic expiry. These usage records do not contain the complete scientific report evidence or generated interpretation. There is no in-app control to delete service records. Canceling generation or deleting a local report does not retract information already sent. Data sent to OpenAI is subject to its applicable data controls and retention policies; see OpenAI data controls. No zero-retention or fixed-deletion guarantee is made for the complete service flow.

CHAPTER 04

Workspace

The application window, Experiment Data capabilities, and Analysis Result capabilities.

Last verified: 2026-09-23

Application window

Selected means the item currently shown in the workspace. **Active** means an experiment is included in operations that use a group of experiments. Selecting a row does not automatically make it Active.

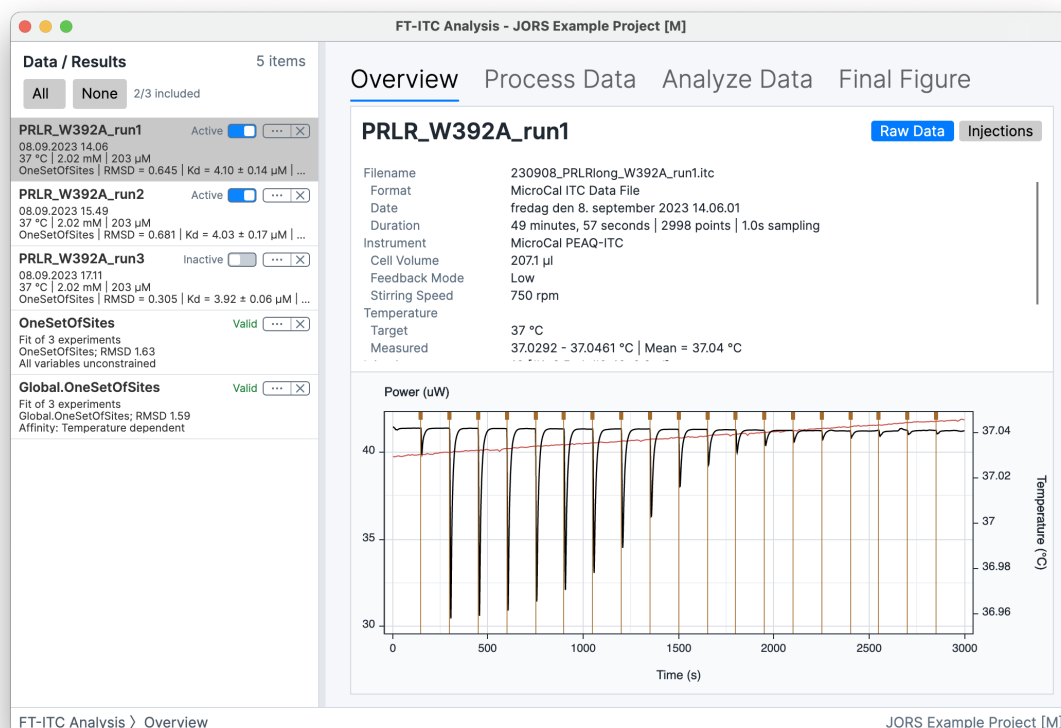
The **Data / Results** list is the project navigator. It contains two types of project item: Experiment Data and completed Analysis Results. Selecting an item shows it in the workspace.

Only Experiment Data has an **Active** toggle. Active experiments participate in operations that use a group of datasets, such as multiple-experiment fitting, processing propagation, and coordinated export. The toggle becomes available after the experiment has been processed. **Enable All**, **Disable All**, and **Invert Active** change several experiments. The **Sort** commands order the list by name, date, temperature, type, ionic strength, or protonation enthalpy. Multiple-experiment fitting is described in Multiple-experiment fitting, and the processing prerequisite is covered in Processing.

The rest of the window changes to match the selected item and task:

- The workflow navigator switches between the available stages of an experiment.
- The workspace contains the graphs, tables, and controls for the current task.
- Workflow controls provide options for the view or analysis currently open.
- Item actions apply to the selected Experiment Data or Analysis Result.
- The application menu contains project commands, tools, preferences, and help.

Experiment Data



Selecting Experiment Data opens its experiment workflow in the shared workspace.

Experiment Data represents an imported or application-created dataset. Its list entry provides identifying information and a summary of its processing or fitted state.

Experiment workflow

Four task views are available for Experiment Data:

- **Overview** presents the imported data and available experiment information.
- **Process Data** turns a raw thermogram into integrated heats through baseline modeling and injection-region control.
- **Analyze Data** configures and runs single- or multiple-experiment fitting.
- **Final Figure** prepares the thermogram, integrated heats, fit, residuals, and annotations for presentation.

Details and attributes

The **Details...** view contains editable concentrations, comments, and experiment attributes. The macOS sheet has separate **Details** and **Attributes** tabs: Details contains the experiment name, date/time, conditions, concentrations, and comments; Attributes contains optional metadata. Attributes describe conditions and analysis inputs such as buffer, salt, ionic strength, competitor, or prebound species.

PLATFORM NOTE (MACOS):

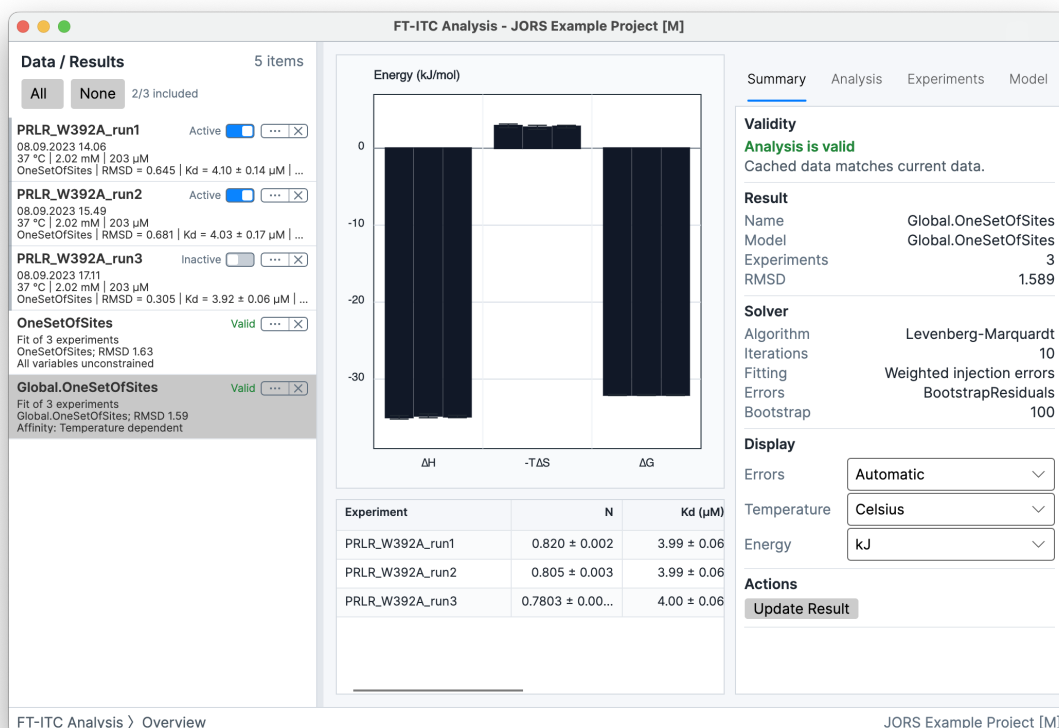
Edit the date and time using separate native controls, including seconds. Use **Add Attribute** on the Attributes tab to add a row. Each tab scrolls independently while **Cancel** and **Apply** remain visible at the bottom. Cancel discards edits; Apply saves them.

Copy Attributes to All copies all attributes from the selected experiment to every other experiment. **Attribute Operations...** copies either one attribute or all attributes to **All other experiments**, **Active experiments**, a **Specific experiment**, or **Experiment names containing...** The name option targets every other experiment whose name contains the entered text, without regard to capitalization. **Clear Attributes** removes all attributes from the selected experiment after confirmation.

Experiment Data actions

- **Details...** opens the editable experiment information and attributes.
- **Duplicate Data** creates another project item from the selected Experiment Data.
- **Export Selected Data...** exports the selected dataset.
- **Clear Solution** removes the fitted solution currently attached to the Experiment Data.
- **Remove Data** removes the item from the open project without deleting its source file.

Analysis Result



An Analysis Result opens its dedicated result workspace when selected.

An Analysis Result stores a fit for one or more experiments together with its model, constraints, fitting settings, uncertainty output, and member solutions. Its list entry summarizes the stored fit.

Result workspace

The result workspace provides these views:

- **Summary** compares fitted and derived parameters across the result and provides its member table.
- **Fit** shows the stored fitted curve and residuals for a selected member experiment.
- **Correlation** shows parameter correlations calculated from residual-bootstrap refits when sufficient bootstrap information is available.
- **Temperature**, **Salt**, and **Protonation** appear when the result and its experiment information support those analyses.

The inspector organizes result information under **Summary**, **Analysis**, **Experiments**, and **Model**. See Results and advanced analyses for interpretation and prerequisites.

Analysis Result actions

- **Details...** edits the result name and comments.
- In Avalonia, **Copy Result Table** copies the selected result table. On macOS, open **Analysis Result Exporter** and click **Copy to Clipboard**. The exporter also provides controlled table export.
- **Update Result** reruns the stored model and fitting settings using the current member experiments and replaces the result only after a successful fit. For bootstrap error estimation, you can increase the number of iterations when updating the result.
- **Set Active Experiments** makes the result's member experiments Active.
- **Load Solutions to Experiments** attaches the stored member solutions to their corresponding Experiment Data.
- **Export Associated Final Figures...** exports one figure per member experiment using that result's saved fit. It loads each saved solution only while making its figure, then restores the experiment's previous solution.
- **Remove Result** removes the Analysis Result from the open project.

Separate tools

The **Tools** menu opens separate windows for tasks outside the main experiment and result workflows. These include **Experiment Designer...**, **Buffer Subtraction...**, and **Experiment Merger...**. Their features are described in Tools.

CHAPTER 05

Processing

Baseline models, integration regions, injection uncertainty, and processing propagation and locking.

Last verified: 2026-09-23

The instrument records power over time, producing a **thermogram** with a peak after each injection. To find the heat from one injection, **Process Data** estimates the background signal (the **baseline**), subtracts it, and measures the area of the remaining peak (integration). It also estimates how much local noise could affect that heat value.

Polynomial and Segmented baselines provide repeatable, model-based treatment with less manual shaping when their assumptions suit the trace. Spline-point editing provides precise local control for difficult cases, while graphical boundary editing, **Fit Peaks**, and copying between injections accelerate integration-region adjustment.

These controls make processing efficient without deciding the scientific interpretation for you. Confirm that the chosen baseline represents the signal between injections and that each integration region captures the observed response.

Raw `.itc`, `.nitc`, `.ta`, and `.apj` imports use this workflow, as do Origin `.opj` imports that contain a usable time/power trace. Integrated-heat imports and Origin projects without a usable trace skip **Process Data**.

CALCULATION:

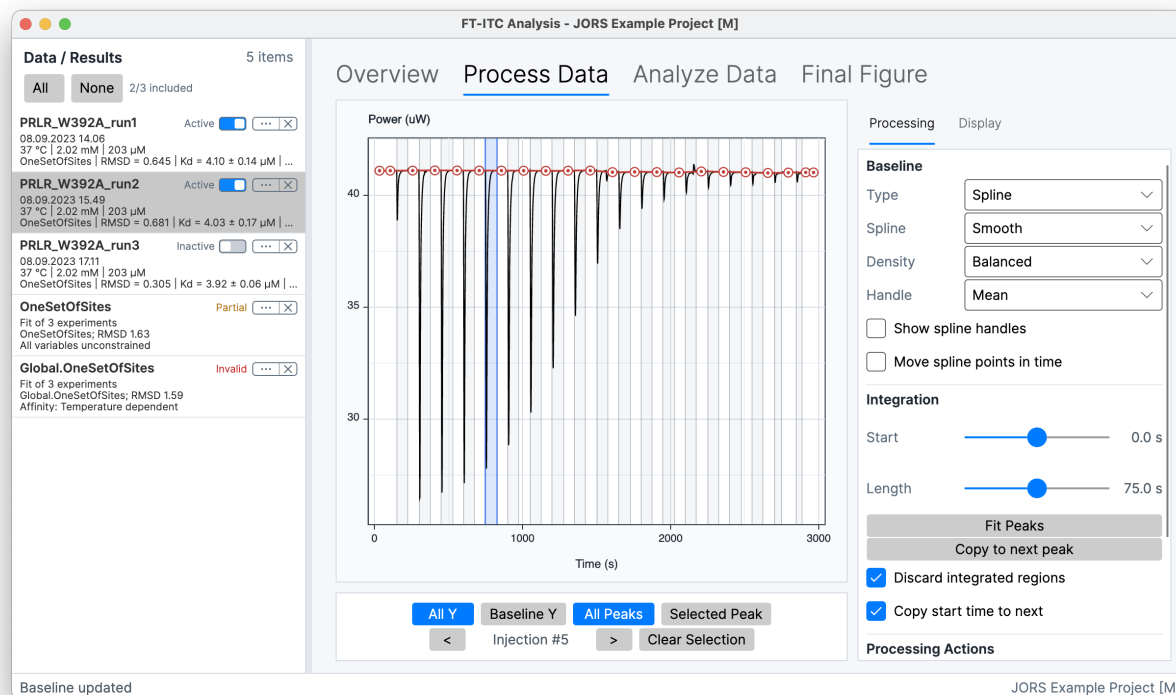
$$q_{i,\text{raw}} = \int [P(t) - b(t)] dt$$

$$\Delta H_i = \frac{q_i}{n_i}$$

$$n_i = c_{\text{syr}} V_i$$

The integral is evaluated between the injection's integration boundaries. Each recorded power value represents the trailing-period average ending at its timestamp, so fractional intervals at either selected boundary are included using the appropriate right-endpoint sample. $P(t)$ is differential power, $b(t)$ is the estimated baseline, and $q_{i,\text{raw}}$ is the raw integrated heat. The heat q_i equals that raw value unless optional buffer subtraction changes it. c_{syr} is the syringe concentration, V_i is the injection volume, n_i is the injected amount, and ΔH_i is the molar heat.

Processing workspace



Process Data combines the thermogram with baseline, integration, display, and navigation controls.

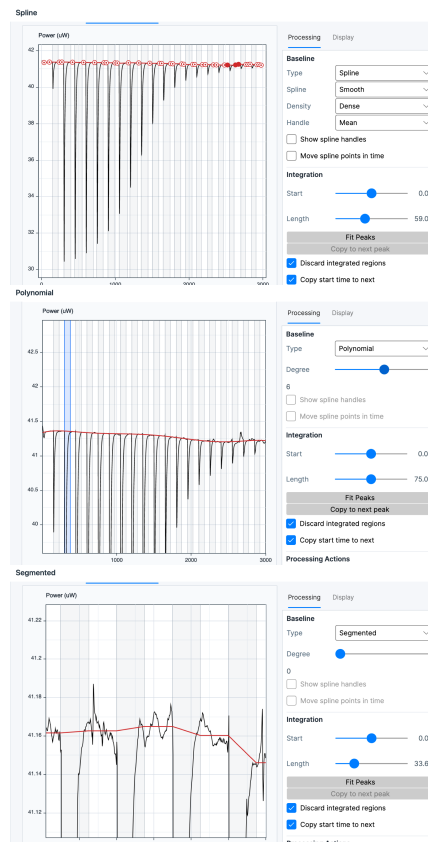
The **Processing** controls configure the baseline and integration regions. The **Display** controls show or hide the baseline, integration regions, corrected data, and cursor information.

With **All injections** selected, **Start** and **Length** apply to every injection. A selected graph region targets one injection, and double-clicking it focuses the graph on that peak. The previous and next controls move through the injections; **Clear Selection** returns to all injections.

The view controls separate horizontal and vertical scaling:

- **All Y** shows the complete power range, while **Baseline Y** emphasizes the baseline region.
- **All Peaks** shows the full injection series, while **Selected Peak** focuses on the current injection.
- Dragging an empty part of the graph zooms into the selected region.

Baseline models and editing



Spline, Polynomial, and Segmented provide different baseline representations for thermogram processing.

Spline

Spline places points in usable baseline regions and interpolates between them. It provides the most direct graphical control.

- **Linear** connects the spline points with straight sections. **Smooth** produces a continuously varying baseline and supports editable slope handles.
- **Sparse**, **Balanced**, and **Dense** change the number of automatically generated points.
- **Mean**, **Median**, and **Min volatility** determine the representative power used for an automatically generated point.
- **Show spline handles** displays the slope handles for a Smooth spline. **Move spline points in time** allows a point to be dragged horizontally as well as vertically.

Dragging a point corrects its position. A secondary-click on the graph adds a point **At Data** or **At Baseline**. A secondary-click on an existing point exposes **Lock** or **Unlock**, **Mark Linear** or **Unmark Linear**, and **Remove**. Marking neighboring points as linear makes the interval between them straight. Locked points are retained when automatic spline points are regenerated.

Balanced provides the middle spline-point density. Greater density increases local flexibility but can also follow noise or absorb part of an injection response.

Polynomial

Polynomial fits one polynomial across the complete thermogram and is suited to smooth global drift. **Degree** controls flexibility.

Polynomial behavior is least constrained at the beginning and end of the run, where a high degree can produce strong edge behavior. Additional degree increases flexibility whether it represents baseline drift or follows more of the trace.

Segmented

Segmented fits local constant, linear, or quadratic baseline behavior between integration regions and blends the local estimates across the run. It is suited to locally changing drift for which one global polynomial is too rigid. **Degree** selects the local behavior.

Local flexibility can follow genuine drift, but it also makes the result more dependent on the integration boundaries and on the baseline immediately around each peak.

Exclude integration regions from the baseline

When **Discard integrated regions** is enabled, data inside the current integration regions are excluded when the baseline is recalculated. Moving a boundary can therefore change both the integrated area and the estimated baseline.

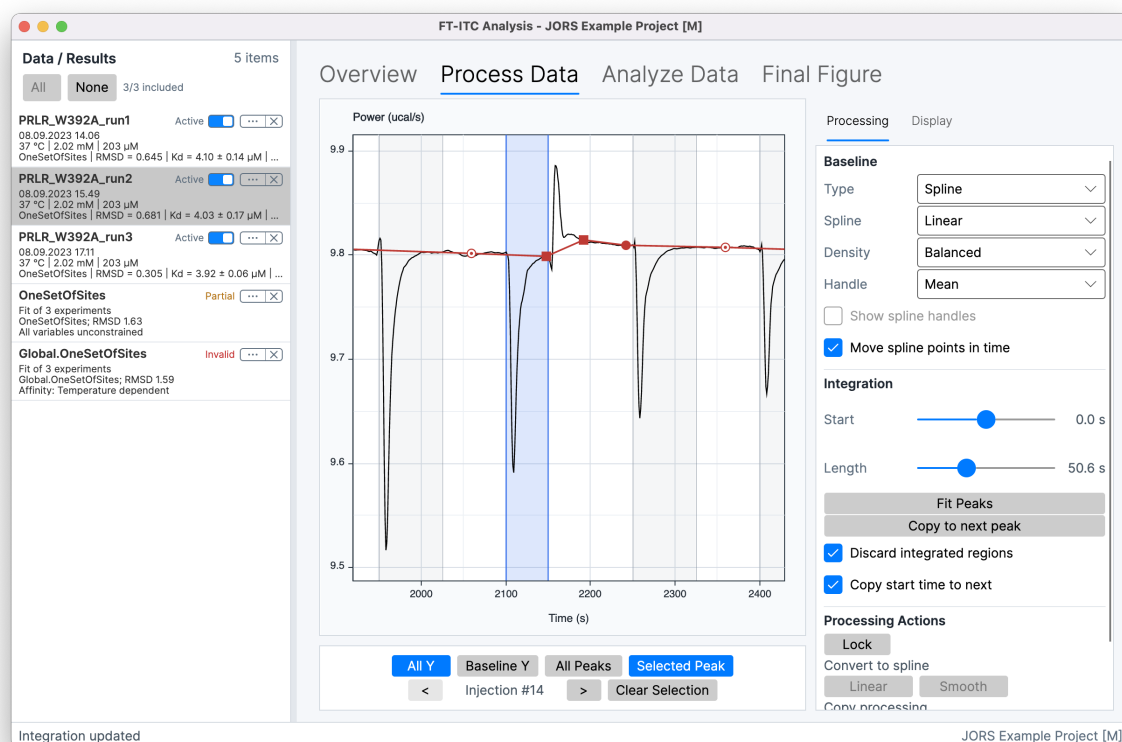
CAUTION:

Boundaries should follow the observed response and a consistent processing rule, not the heat expected from a model.

Convert to a spline

A Polynomial or Segmented baseline can be converted to a **Smooth** or **Linear** Spline when the automatic baseline is a useful starting point for graphical editing. Conversion changes the baseline representation and creates editable spline points.

Integration regions



Selecting an injection exposes its individual boundaries and navigation controls.

Each injection region has a start and an end boundary. **Start** sets the offset of the start boundary relative to the injection. The value displayed as **Length** positions the end boundary that many seconds after the injection begins; it is not a separate processing mode.

Either boundary can be dragged in the graph or adjusted with the controls. The application constrains the start and end to a valid interval within the injection scope and preserves a minimum separation between them.

Estimate end points with Fit Peaks

Fit Peaks estimates the end point of each injection from the decay of the baseline-corrected response. It changes the end boundaries and then integrates the resulting regions; it does not fit the integrated heats or select a persistent integration mode.

When peak fitting converges, the estimated boundaries replace the previous end points. If fitting fails or does not converge, the previous regions remain unchanged. Peak kinetics can vary across the titration, and the first injection can behave differently from the remaining series.

Copy a region to the next injection

Selecting an injection and choosing **Copy to next peak**, or pressing **Space**, copies its end boundary and advances to the next injection. **Copy start time to next** includes the start boundary in that operation.

Injection uncertainty

Each injection error bar shows an estimated ± 1 standard deviation for that injection's molar heat: roughly, how much the calculated heat might vary because of local noise in the thermogram. It is calculated independently for every injection, so the bars can vary across a titration. It does not cover every source of experimental error.

The calculation uses baseline-corrected samples around the injection. It combines an estimate of local power noise with the temporal correlation between neighboring samples, the integration-region length, and uncertainty in the baseline level. A longer or noisier region will therefore often have a larger estimated uncertainty. When Buffer Subtraction is applied, the independent target and reference heat uncertainties are combined.

The correlation estimate uses only consecutive baseline sample pairs separated by at most twice the sampling interval. Both its product sum and its normalization use those same pairs, excluding a pair across a longer integration gap. Propagation assumes an AR(1) noise model, in which correlation decays geometrically with sample separation; this is a chosen approximation to the local noise.

Previously processed projects can retain heat SDs calculated with a normalization that included an unpaired sample across the gap. Reintegrate affected thermograms to update those SDs, then rerun affected weighted fits and parameter-uncertainty calculations before updating published tables or figures. These processing SDs supply fit weights; the weighted objective and displayed unweighted RMSD remain separate quantities.

NOTE:

If an injection has no processing-derived uncertainty estimate, its stored error is zero and no error bar appears. Integrated-heat imports have no thermogram from which to calculate this estimate. A zero or absent error bar does not establish that the injection has no uncertainty.

The bars do not include uncertainty in cell or syringe concentration, fitted parameters, or model-derived confidence bands. They also do not quantify baseline-model choice, integration-boundary choice, calibration error, other instrumental effects, or other systematic uncertainty. Consequently, they do not validate the selected processing or constitute confidence intervals for the true heat.

Weight by injection error can use these estimates in the fitting objective. Concentration uncertainty is handled separately in supported resampling calculations. Both behaviors are described under Fitting calculation.

Propagate and lock processing

Active under **Copy processing** copies the selected experiment's processing to the other Active experiments. This can replace existing processing when a destination is unlocked. **New** targets experiments that do not yet have processing.

Processing can be propagated while the source is unlocked. Locked destinations are not overwritten. After propagation, each processor that should be protected can be selected and set to **Lock**. Locking disables baseline, spline-point, integration-region, and peak-fitting edits; **Unlock** makes changes available again.

CHAPTER 06

Single-experiment fitting

Fit one experiment, configure model and uncertainty options, control injection inclusion, and interpret fit diagnostics.

Last verified: 2026-09-23

After processing a thermogram or importing integrated heats, you have one heat value per injection. A binding model predicts those heats from quantities such as affinity and binding enthalpy. **Analyze Data** adjusts the model parameters to bring its predicted heats close to the observed heats. In **Single experiment** mode, it uses the included injections from the selected Experiment Data, along with the concentrations, injection volumes, cell volume, temperature, and any model-specific information.

The fitted values describe this dataset under the selected model. Check the pattern of differences between prediction and observation (**residuals**) and whether the fitted values make scientific sense; a successful optimizer stop alone is not a model check.

The inspector has four tabs:

- **Fit** selects the model, optimizer, error-estimation method, limits, weighting, and result output.
- **Parameters** shows the model parameters and their starting or fixed values.
- **Options** contains settings specific to the selected model.
- **Display** controls the fitted curve and diagnostic information shown in the graph.

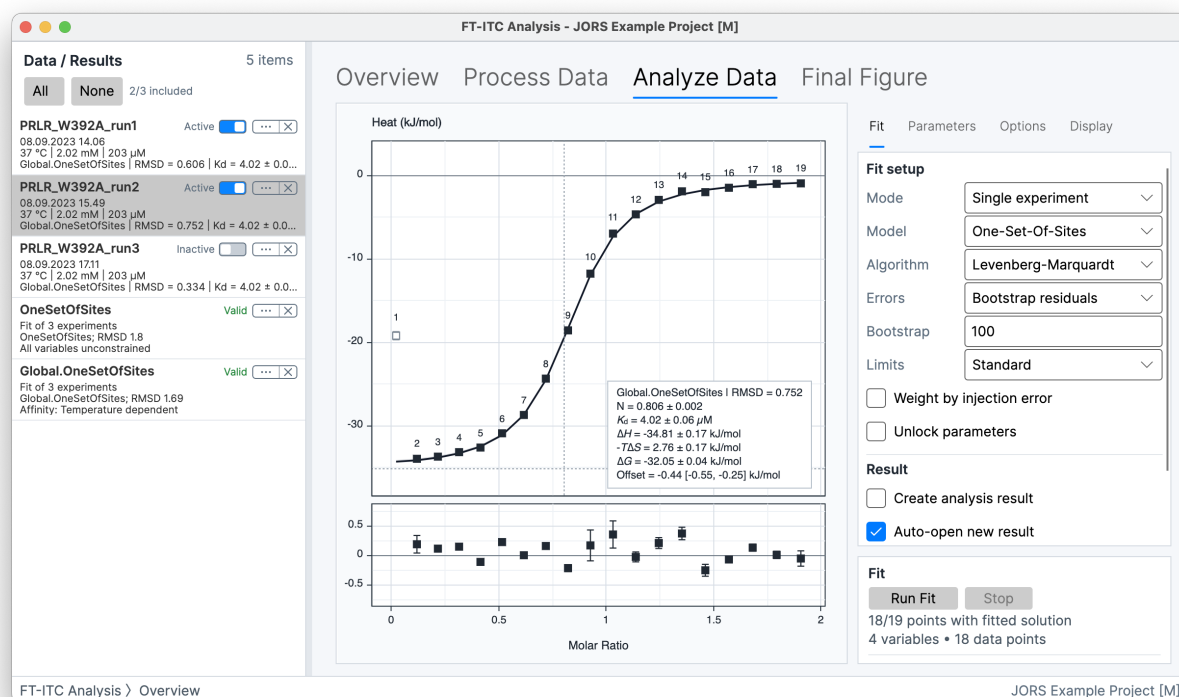
Profile-likelihood uncertainty

Profile likelihood estimates parameter uncertainty by asking how far a fitted parameter can be moved away from its best-fit value before the agreement with the experimental data becomes significantly worse. Profile likelihood does not create a refit ensemble or a confidence band.

For each fitted parameter, FT-ITC first starts from the best-fit solution. The parameter of interest is then fixed at a different value, while all other fitted parameters are allowed to readjust. This is repeated in both directions from the best-fit value. The resulting profile therefore accounts for compensation between parameters: for example, a change in affinity may partly be compensated by changes in enthalpy, stoichiometry, or other fitted parameters.

The 95% confidence limits are the parameter values at which the best achievable fit reaches a statistically defined 95% threshold. FT-ITC uses an F-calibrated threshold based on the number of observations and fitted parameters. The confidence interval is therefore determined by the shape of the fit surface rather than by assuming that parameter uncertainty is symmetric or normally distributed.

A complete interval is reported only when the threshold is crossed on both sides of the best-fit value. If the search reaches a parameter limit first, that side is recorded as censored and has no confidence endpoint. For a complete interval, the displayed `value ± SD` uses a symmetric display equivalent calculated from the asymmetric interval; it is not a sample standard deviation. Save the result in an `.ftxtc` project to preserve the profile diagnostics.



Analyze Data combines the fitted-heats graph and residuals with the controls for configuring and running the fit.

Multiple-experiment fitting uses additional experiment selection and parameter constraints; see Multiple-experiment fitting.

Injection inclusion

Selecting an injection point in the integrated-heats graph changes whether it is included in the fit. Excluded injections do not contribute to the objective function.



Excluded injections remain visible when Excluded points is enabled and can be selected for inclusion again.

Changing injection inclusion does not rerun the fit. The fitted curve and parameters continue to represent the previous fit until **Run Fit** is used again. An inclusion change can also invalidate an Analysis Result that contains the experiment.

Injection bookkeeping: MicroCal, Ideal continuous mixing, and Discrete displacement

Injection bookkeeping determines the concentrations used by the fit and accounts for reaction heat carried out of the active cell by displaced solution. It does not change measured peak areas or baseline integration.

- **MicroCal** uses the dilution and displaced-heat approximations documented for MicroCal analysis. Use it to reproduce or compare results calculated with that convention.
- **Ideal continuous mixing** models concentrations with ideal exponential mixing and accounts for displaced heat along the continuous mixing trajectory. It uses the exponential concentration law previously exposed as the Exponential preference, but its heat calculation is different. This is the convention previously labelled Dumas in the user interface.
- **Discrete displacement** is the default and recommended starting point for ordinary pulse injections. It represents the physical limiting case where injection displaces the previous cell mixture before appreciable mixing, then the remaining material mixes with the injected

solution. This follows the discrete-injection formalism described by Freire, Schön and Velazquez-Campoy (2009), also implemented by pytc. FT-ITC retains its own equilibrium solvers.

For **MicroCal**, let u be cumulative injected volume divided by active cell volume, M_0 be the initial cell concentration, and C_s the syringe concentration. Starting with no ligand in the cell:

CALCULATION:

$$M = M_0 \frac{1 - \frac{u}{2}}{1 + \frac{u}{2}}$$

$$X = C_s u \left(1 - \frac{u}{2}\right)$$

The first line estimates how much macromolecule remains; the second estimates how much injected ligand is in the cell.

These equations estimate the macromolecule concentration remaining in the cell (M) and the ligand concentration from the syringe (X) after the injections. They describe the **MicroCal** bookkeeping choice; they are not measured curves or a separate tool.

Every delivered injection contributes to cumulative volume, including injections excluded from fitting. For a tandem experiment, this calculation stops when cumulative injected volume reaches twice the active cell volume; ideal continuous mixing has no corresponding cumulative-volume limit. Existing projects retain saved concentrations and fits until concentration reprocessing.

For one injection, let v be injection volume, V active cell volume, and Q the equilibrium binding heat content of the cell in joules. The predicted injection heat q depends on the bookkeeping method.

MicroCal:

CALCULATION:

$$q = Q_{\text{end}} - Q_{\text{start}} + \frac{v}{V} \frac{Q_{\text{start}} + Q_{\text{end}}}{2}$$

The heat change is adjusted for displaced solution using the average heat content before and after the injection.

Ideal continuous mixing:**CALCULATION:**

$$q = Q_{\text{end}} - Q_{\text{start}} + \frac{v}{V} \frac{Q_{\text{start}} + 4Q_{\text{mid}} + Q_{\text{end}}}{6}$$

This averages displaced heat along the mixing path, including the state halfway through the injection.

Discrete displacement:**CALCULATION:**

$$q = Q_{\text{end}} - \left(1 - \frac{v}{V}\right) Q_{\text{start}}$$

Only the fraction of the original cell mixture left after displacement contributes its starting heat.

For **Discrete displacement**, concentrations advance after each injection as follows:

CALCULATION:

$$M_{\text{end}} = \left(1 - \frac{v}{V}\right) M_{\text{start}}$$

$$X_{\text{end}} = \left(1 - \frac{v}{V}\right) X_{\text{start}} + \frac{v}{V} C_s$$

The same fraction of each starting concentration remains, while the injected solution adds ligand from the syringe.

Across a fixed-syringe segment, multiply the fraction retained after each injection; do not use an exponential of cumulative volume. Every shot must be smaller than the cell volume; cumulative injected volume can exceed it. Zero-volume steps leave the state unchanged. No numerical integration or substeps are used. Displaced heat is treated consistently with this discrete replacement: the correction uses the pre-injection heat content, rather than MicroCal's average of the start and end heat contents.

The midpoint is evaluated halfway through the injection on the exponential concentration trajectory. Dissociation additionally accounts for heat carried by associated complex entering from the syringe; its legacy calculation is retained under MicroCal. Offsets are applied separately, as before.

Preferences > Processing > Injection bookkeeping selects the default for new data.

Experiment Details > Injection bookkeeping explicitly switches an ordinary experiment and invalidates its fits without reintegrating measured heats. Changing the preference or editing a name/comment does not switch existing data. Older projects retain their saved concentrations and historical heat behavior; if their method is unknown, the selector displays **Saved processing — unchanged**. Choose a method explicitly before recalculating unknown saved concentrations. Rebuild tandem experiments through the tandem tool to change their method.

The ideal continuous mixing convention is the opposite physical limit: it assumes mixing throughout the injection, so the outgoing mixture changes composition continuously. Consider it for injections slow relative to cell mixing or as a sensitivity comparison with the discrete limit. It is inspired by Dumas (2022), with an FT-ITC finite-injection numerical integration; it does not implement that paper's imperfect-mixing/adjustable-volume model or kinetic single-injection analysis. No option is assumed to be empirically superior. Simpson integration uses one fixed panel per injection, so unusually large injections or very sharp transitions can need additional scrutiny; there is no adaptive refinement. It requires three equilibrium states, whereas MicroCal and Discrete displacement require two.

Discrete displacement is useful when comparing analyses that use the same injection convention. It is not an imperfect-mixing correction. FT-ITC retains its own equilibrium solvers and offset convention, so results may differ from other programs even when the same displacement bookkeeping is selected. Fractional-stoichiometry two-site binding and the Dissociation model use FT-ITC-specific extensions of the convention. Neither fitted parameters nor background-heat conventions are automatically converted between programs.

Models

Choose a model based on what is in the cell and syringe and what interactions are plausible. Adding parameters can improve a curve's fit even when the extra binding process is unsupported, so compare residuals and parameter uncertainty as well as the curve. The descriptions below state what each model assumes and reports.

One-Set-Of-Sites

One-Set-Of-Sites represents one class of equivalent, independent binding sites. It fits stoichiometry, dissociation constant, binding enthalpy, and an injection-heat offset. The solution also reports thermodynamic quantities derived from the fitted affinity and enthalpy.

Let M be the total macromolecule concentration, N the number of sites per macromolecule, X the total ligand concentration, x the free ligand concentration, and K_d the dissociation constant. A smaller K_d means stronger binding. The model finds x by balancing free and bound ligand, then calculates the heat content Q of the cell:

CALCULATION:

$$f = x / (K_d + x)$$

$$X = x + N M f$$

$$Q = V N M f \Delta H$$

The fraction f tells us how many sites are occupied. The middle line counts free ligand plus ligand on the sites. The last line multiplies the amount bound in cell volume V by the heat per mole bound, ΔH . Injection heat then follows the selected bookkeeping calculation above.

The **Options** tab provides **Use Syringe Correction** and **Stoichiometry**. Without syringe correction, the fitted N-value represents the apparent site stoichiometry. With syringe correction enabled, **Stoichiometry** fixes the number of cell-side sites and the fitted N parameter becomes the active syringe-concentration factor `alpha`.

The model cannot by itself distinguish a concentration error from other effects that change an apparent stoichiometry.

Two-Sets-Of-Sites

Two-Sets-Of-Sites represents two independent classes of sites. It fits separate stoichiometries, dissociation constants, and enthalpies for the two classes, together with a shared injection-heat offset.

Each class has its own site count N , affinity K_d , and binding enthalpy ΔH . With free ligand concentration x , the two classes share the same supply of ligand:

CALCULATION:

$$f_j = x / (K_{d,j} + x), \text{ for } j = 1 \text{ or } 2$$

$$X = x + M(N_1 f_1 + N_2 f_2)$$

$$Q = V M(N_1 f_1 \Delta H_1 + N_2 f_2 \Delta H_2)$$

Each f is the occupied fraction for one site class. The ligand balance finds one free concentration that satisfies both classes. Their heat contributions add to give the cell heat content; injection bookkeeping converts its change to a predicted heat.

Shared N-Values makes the two site classes use the same fitted stoichiometry. **Use Syringe Correction** instead fixes the first and second **Stoichiometry** values and fits one active syringe-concentration factor, `alpha`.

The two site labels are interchangeable: exchanging all parameters assigned to site 1 and site 2 describes the same physical model. A lower RMSD or optimizer objective alone does not establish that two distinguishable binding processes are supported by the experiment.

Sequential Binding Sites

Sequential Binding Sites represents two, three, or four ordered binding steps on a macromolecule in the cell. **Sequential binding steps** in the **Options** tab selects the number of steps. The model fits one macroscopic stepwise association constant and one molar step enthalpy for each transition, together with the ordinary molar injection-heat offset. It does not fit an N-value or syringe activity.

For step count n , let $\beta_0 = 1$, $\beta_i = \prod_{j=1 \dots i} K_j$, and let x be the free ligand concentration. The state weights and fractions are

CALCULATION:

$$w_i = \beta_i x^i$$

$$F_i = \frac{w_i}{\sum_{j=0}^n w_j}$$

$$\bar{\nu} = \sum_{i=0}^n i F_i$$

$$X_t = x + M_t \bar{\nu}$$

Each state receives a weight from its binding steps and the available free ligand. Dividing by the total weight gives the fraction in that state. The average number of bound ligands then balances free and bound ligand against the total.

Here M_t and X_t are total macromolecule and ligand concentrations in the cell. The model solves the ligand balance as part of the fit and calculates the cell heat content from the population of every sequential state:

CALCULATION:

$$Q = VM_t \sum_{i=1}^n F_i \left(\sum_{j=1}^i \Delta H_j \right)$$

For each state, add the heats of the steps needed to reach it, multiply by the fraction in that state, and sum across states. Multiplying by the macromolecule amount in the cell gives its heat content.

The reported K_i values are phenomenological, macroscopic step constants for the ordered transitions $M \rightarrow MX \rightarrow MX_2$ and so on. They are not microscopic intrinsic site constants. Step numbers therefore have a physical order, so fitted steps are never sorted or treated as interchangeable.

The macromolecule must be in the cell and ligand in the syringe. Reverse titrations with macromolecule in the syringe are outside this model. Multi-step fits can be weakly identifiable, especially when the concentration window does not populate every transition. Inspect parameter bounds, residuals, bootstrap uncertainty, and parameter correlations; an improved RMSD does not by itself establish the selected number of sequential steps.

Competitive Binding

Competitive Binding represents titration of a target ligand into a macromolecule that is initially in equilibrium with a prebound ligand in the cell. It fits the target ligand's stoichiometry, dissociation constant, binding enthalpy, and injection-heat offset.

The target (A) and competitor (B) compete for the same sites. Let a and b be their free concentrations, $K_{a,A}$ and $K_{a,B}$ their association constants, and S the total site concentration. The model solves for the two free concentrations so that the available ligand is accounted for:

CALCULATION:

$$D = 1 + K_{a,A}a + K_{a,B}b$$

$$f_A = K_{a,A}a / D; f_B = K_{a,B}b / D$$

$$A_{\text{total}} = a + S f_A; B_{\text{total}} = b + S f_B$$

$$Q = V S(f_A \Delta H_A + f_B \Delta H_B)$$

D counts the empty and occupied possibilities for a site; f_A and f_B are the occupied fractions. The ligand balances count free plus bound molecules. Each bound population contributes its own heat. The competitor properties are supplied; the target properties are fitted.

The **Options** tab requires the pre-equilibrated competitor's **Total competitor** concentration, **Ligand Affinity**, and **Ligand Enthalpy**. **Total competitor** is the total analytical competitor concentration in the cell after pre-equilibration: free competitor plus competitor bound to the macromolecule. Do not enter only the initially bound complex. **From attributes** makes **Total competitor** use the corresponding value stored in the Experiment Data attributes instead of the value entered in the model options. **Ligand Affinity** and **Ligand Enthalpy** each have a separate **From attributes** option. Add a **Competitor properties** experiment attribute and select a one-set-of-sites Analysis Result to supply its global summary Kd and ΔH , including their uncertainty. Each experiment in a global fit can select its own result. The row shows the source status; hover over it or the selector for the full result name and a short Kd and ΔH summary with units and SD. A missing source can still use captured values when available. Opening the editor only previews values; the saved capture refreshes when a fit starts. The model also provides **Use Syringe Correction** and **Stoichiometry** with the same concentration-factor interpretation as One-Set-Of-Sites.

Model options remain available to edit even when no experiment is ready for fitting. When **From attributes** is selected, starting a fit copies each experiment's attribute value into its model option; the entered value is used only if **From attributes** is turned off. Different experiments in a global fit can therefore have different effective option values. Every included experiment must have the attributes required by the enabled selections. If any are missing, one error dialog lists the affected experiments and attributes so they can be added or the selections turned off. A fit that cannot start leaves any previously attached solution in place. **Update Result** reads the attributes again; existing results continue to use the values saved with their fits.

The **Total competitor** experiment attribute has separate concentration and SD fields in both desktop apps. Buffer, salt, and ionic strength concentration attributes have a value field without an SD field.

If the source result has temperature dependence enabled, its summary is evaluated at the experiment temperature. Otherwise its usual summary temperature is used, with a notice when the experiment temperature differs. The attribute keeps the values used by the fit: if the source

result is removed, existing results remain valid and a new fit can use the captured values. A source removed before any values were captured must be restored or replaced before fitting. Updating the source result marks dependent analysis results stale; fitting again copies the updated summary. Residual bootstrap samples the copied affinity and enthalpy uncertainties separately, as it does for manually entered model options.

The **Ligand Affinity** and **Ligand Enthalpy** labels describe the pre-equilibrated competitor's properties. The fitted target affinity and enthalpy depend on those supplied properties. The competitor properties are model inputs, not quantities independently determined by this fit.

The reported apparent target K_d includes a competition factor calculated from the initial free competitor concentration. This calculation accounts for competitor already bound to cell sites rather than substituting the total competitor concentration. The apparent target K_d can therefore differ from its intrinsic value.

Dissociation

Dissociation represents an associated species separating into two free components as it is diluted. This includes a 1:1 complex of two different components (a heterodimer) when they are present in equal amounts. The syringe contains the associated species in equilibrium with its free components, and the cell initially contains buffer. Dilution changes the fraction associated. The model fits an effective dissociation constant, the heat per mole of associated species formed, and an injection-heat offset.

CALCULATION:

$$K_{a,model} = P / F^2$$

$$C = F + 2P$$

$$Q = V P \Delta H_{assoc}$$

F is the concentration of free components counted together, P is the concentration of associated pairs, and C counts two components per pair. The equilibrium determines how much remains associated at each dilution. The cell heat content is the amount of associated pair times the heat of forming it. Dilution drives pairs apart, so dissociation heat has the opposite sign from the fitted association enthalpy. Injection bookkeeping also accounts for associated species carried in from the syringe.

For an equimolar complex $A + B \rightleftharpoons AB$, enter the syringe concentration as the **sum of the concentrations of A and B units** (twice the concentration of an equimolar AB preparation). Then $F = [A] + [B]$ and $P = [AB]$. Because $[A] = [B] = F/2$, the usual heterodimer association constant $K_{a,AB} = [AB]/([A][B])$ is **four times** the model's $K_{a,model}$; the usual heterodimer K_d is one quarter

of the fitted K_d . The fitted ΔH_{assoc} is per mole of AB formed. The model does not represent unequal component amounts, separate component concentrations, or other oligomerization schemes. It has no stoichiometry or syringe-correction options.

Thermodynamic relationships

The application derives thermodynamic quantities from the fitted affinity and enthalpy. Temperature T is expressed in kelvin.

CALCULATION:

$$\begin{aligned}K_d &= 1/K_a \\ \Delta G &= RT \ln(K_d) = -RT \ln(K_a) \\ -T \Delta S &= \Delta G - \Delta H\end{aligned}$$

Here, R is the gas constant, and the final relationship matches the **-TΔS** quantity reported by the application.

Parameters and model options

Fit
Parameters
Options
Display

N-value
N
☐ Locked

unitless

Enthalpy
ΔH
☐ Locked

kJ/mol

Affinity
K_d
☐ Locked

μM

Offset
Offset
☐ Locked

kJ/mol

Fit
Parameters
Options
Display

Use Syringe Correction
☐ Enabled

Use a syringe concentration correction factor instead of fitting an apparent N-value. Useful when the active titrant concentration is uncertain.

Stoichiometry

Fixed stoichiometric site ratio used by the model. Controls how many binding sites are represented on the cell side.

The application generates initial parameter values from the experiment and can reuse an attached fitted solution where applicable. Entering a value replaces the generated starting value for that parameter. Values are displayed in the current application units. Clearing a value returns the parameter to automatic initialization.

Locked holds a parameter at its displayed value during the primary fit. An unlocked parameter is adjusted by the optimizer. Locked values remain part of the model and affect every other fitted parameter even though they are not estimated by that fit.

Analysis choices are retained separately for the available fitting modes and models. **Restore defaults** clears the stored analysis inputs and reloads the fitting controls from the current preferences. It also resets the inspector's **Unlock parameters** setting. The action does not change preferences, including saved limits, result-output options, and display settings.

The **Limits** control selects a common parameter-bound policy:

- **Standard** uses the normal parameter bounds.
- **Expanded** permits a wider parameter range.
- **No limits** removes the configured parameter bounds.

A fitted value at a bound is not an interior estimate. It indicates that the reported value depends on the selected bound as well as on the data and model.

Before a fit starts, the application checks every free starting parameter against the currently selected limits. This includes values entered manually, values reused from an attached solution, and automatic values that become stale after the **Limits** policy changes. Bounds are inclusive, so a value exactly at a bound is allowed. Locked and globally determined parameters are not checked because they are not optimizer coordinates. An out-of-range value is marked in the **Parameters** inspector and **Run Fit** remains available so it can be corrected. The fit is blocked with a list of affected parameters; edit the value, clear it to restore the automatic default, or widen **Limits**. In a global fit, a local parameter that has no exposed global editor is identified by its experiment name in the fit status area; edit that experiment in single-experiment mode if needed.

Fitting calculation

Algorithm

Levenberg-Marquardt uses local derivative information and can be efficient when the starting values describe a suitable region of the fitting surface.

Nelder-Mead is a derivative-free simplex optimizer. It provides an alternative when the derivative-based method has difficulty converging from the chosen starting values.

Weight by injection error

Weight by injection error uses the integration uncertainty estimated during thermogram processing when calculating the fitting objective. An injection with a larger estimated error has less influence on the fitted parameters than one with a smaller estimated error. Choose this only when those processing-derived error estimates are suitable for the dataset.

CALCULATION:

$$r_i = q_{i,\text{obs}} - q_{i,\text{model}}$$

$$\text{RMSD} = \sqrt{\left[\frac{\sum r_i^2}{N} \right]}$$

$$\text{weighted objective} = \sum (r_i / \sigma_i)^2$$

Only included injections enter these sums. The value σ_i is the processing-derived uncertainty for injection i ; weighting changes the fitting objective but does not remove systematic uncertainty.

RMSD summarizes the typical size of the heat differences between the observed points and the fitted curve. It is always calculated from the unweighted residuals, including after a weighted fit. The result tooltip reports the optimizer objective separately: the raw residual sum of squares for unweighted fitting or the standardized residual sum of squares for weighted fitting. Weighting can change the fit even though the displayed RMSD remains an unweighted summary.

For a multiple-experiment result, the displayed global RMSD is pooled across every included injection in every member experiment. Each member row retains its own local RMSD, so the global value remains comparable when members contain different numbers of included injections.

Weight by injection error is available only when every included injection has a finite, positive peak-area SD. The application checks this again before fitting. An injection without an estimated SD can be excluded from the fit, or its thermogram can be reprocessed to obtain an estimate.

The weighting describes the application's processing-derived uncertainty model. It does not account for every systematic source of experimental or processing uncertainty.

Parameter uncertainty

The reported parameter value comes from the best fit to the original data. The **Errors** control chooses how to estimate its uncertainty afterward; these methods do not replace that reported value with an average of refits:

- **None** retains the primary fit without resampling-based parameter uncertainty.
- **Bootstrap residuals** samples from the included injections' centered primary-fit residuals and adds each draw to a best-fit prediction. For an unweighted fit, it uses the raw residuals: injection peak-area SDs do not affect the synthetic peak areas or refits. This assumes the residual errors have roughly equal spread across injections. For an error-weighted fit, it divides each residual by its peak-area SD before centering and sampling, then multiplies each draw by the target injection's SD. Synthetic injections retain their stored peak-area SDs, so weighted refits use the same per-injection weights.
- **Leave-one-out** performs one deterministic refit for each deletion: one refit per included injection in a single-experiment analysis, or one refit per omitted experiment in a globally fitted multiple-experiment analysis. Concentrations, uncertain model options, and parameter locks are held at their primary-fit values so the resulting spread isolates deletion sensitivity.
- **Profile likelihood** fixes each fitted coordinate in turn and refits nuisance coordinates against the conditional likelihood. It uses the local objective for independent members and the complete objective for shared global coordinates; complete asymmetric crossings are retained as endpoints, while a bound reached before crossing is reported as censoring.

Bootstrap sets the requested number of residual-bootstrap iterations. It is enabled and used only for residual bootstrap; leave-one-out and profile likelihood have deterministic schedules and do not use this count. Only included injections supply residuals, and only retained usable refits enter the parameter distributions. Because residual-bootstrap sampling is with replacement, one residual can occur more than once in a synthetic dataset while another may not occur at all. The fit status distinguishes successful and failed refits.

Correlation diagnostics distinguish attempted refits, optimizer-usable refits, and the listwise-complete refits whose displayed coordinates are all finite. A high failed-refit fraction can make the retained ensemble selective; increasing the requested count improves Monte Carlo resolution but does not correct a systematically failing refit process.

Each replicate uses a fresh independent random stream; seeds are not stored, so rerunning a bootstrap does not reproduce the same random sequence.

When **Update Result** is used on a stored residual-bootstrap Analysis Result, its dialog shows the retained usable-refit count and offers the stored behavior plus larger supported presets up to 10,000 requested iterations. The update performs a fresh complete fit and bootstrap; it does not append samples to the saved distribution. Cancelling the calculation or completing it without any usable bootstrap refits preserves the previous Analysis Result.

The primary best-fit parameter remains the reported value. For a parameter with best-fit value $\hat{\theta}$ and values θ_b from B retained refits, the application summarizes the bootstrap distribution as follows:

CALCULATION:

$$SD = \sqrt{\frac{\sum_b (\theta_b - \hat{\theta})^2}{B}}$$

$$95\% \text{ CI} = [P_{2.5}(\{\theta_b\}), P_{97.5}(\{\theta_b\})]$$

SD is the root-mean-square deviation of the retained refits from the primary best fit. The confidence limits are the 2.5th and 97.5th percentiles of the retained refit distribution itself, so they need not be equally spaced around the best fit. Neither calculation replaces the best-fit value with the bootstrap mean or median.

The uncertainty display can show SD, the 95% confidence interval, both, or select between them automatically. This presentation rule is described under Uncertainty and evaluation temperature.

Concentration uncertainty

With **Concentration uncertainty** enabled in the **Fit** tab, the concentration SDs entered in **Details...** are propagated through residual-bootstrap calculations. The control is initialized from the corresponding preference and is active only for residual bootstrap; leave-one-out and profile likelihood keep primary concentrations fixed. Each nonzero fractional SD is the arithmetic standard deviation relative to the entered concentration. Synthetic clones draw a positive, mean-preserving lognormal multiplier: if the fractional SD is c , then $\sigma^2_{\log} = \ln(1 + c^2)$, $\mu_{\log} = -\sigma^2_{\log}/2$, and the multiplier is $\exp(\mu_{\log} + \sigma_{\log}Z)$ for a standard-normal Z . Thus the multiplier has mean 1 and SD c , so cloned concentrations remain positive while preserving the entered arithmetic mean and SD. Explicit cell or syringe SDs take precedence over the automatic value configured in Preferences. These uncertainties affect the synthetic experiment concentrations used for bootstrap refits, not the concentrations used for the primary best fit.

Displayed parameter uncertainty

The bootstrap summary is first calculated for each fitted parameter coordinate. The application then converts that summary into the quantity shown to the user. For example, affinity is fitted as $\log_{10}(K_d)$ but is normally displayed as K_d . The displayed central value comes from the primary best fit, SD is propagated through the transformation, and the percentile limits are transformed and reordered as required.

Quantities calculated from more than one reported parameter, such as $-T\Delta S$, use the application's uncertainty-propagation rules for that calculation. Their displayed limits are therefore not necessarily the percentiles that would be obtained by recalculating the complete derived quantity independently for every bootstrap refit. The **Automatic** SD-or-CI decision is applied after transformation or propagation, separately for each displayed quantity.

Unlock parameters during error estimation

Locked parameters remain fixed during the primary fit. With **Unlock parameters** enabled, copies of those parameters are unlocked only for residual-bootstrap refits and can vary in the resampled solutions. Leave-one-out and profile likelihood always preserve the primary-fit locks.

This setting does not change or rerun the primary best fit. It changes only the parameter state used by the repeated error-estimation fits, and it has no effect when no fitted parameter is locked.

Fit execution and diagnostics

Run Fit starts the primary optimization and the selected error-estimation calculation. **Stop** requests cancellation of the active calculation. When fitting ends, the status reports the termination state, RMSD, iteration count, and elapsed time. A resampling calculation also reports its outcome and the number of successful and failed refits.

The **Display** tab exposes complementary diagnostics:

- **Fit line** shows the curve calculated from the current fitted solution.
- **Residuals** show the difference between each included observation and the fitted curve.

- **Error bars** show the processing-derived integration uncertainty.
- **Confidence band** shows uncertainty around the fitted curve when the solution contains suitable resampling results.
- **Excluded points** shows injections that do not contribute to the current fit.

These displays describe different aspects of the fitted solution. A small RMSD does not rule out systematic residual structure, poorly identified parameters, or dependence on model assumptions.

Store the fitted solution

With **Create analysis result** disabled, a successful single-experiment solution remains attached to its Experiment Data and is available in that experiment's analysis and figure workflows.

With **Create analysis result** enabled, a usable completed fit also creates a separate Analysis Result containing the experiment, model, fit settings, and solution. **Auto-open new result** determines whether the new result workspace opens immediately. A stopped or unusable fit does not create or replace an Analysis Result.

Fit availability and non-convergence

A single-experiment analysis requires processed or imported heats and at least three included injections with usable numerical values. The binding models also require nonzero cell and syringe concentrations. **Dissociation** uses the syringe concentration and does not require a macromolecule concentration in the initially buffered cell.

An unavailable model, failed termination, bound-limited solution, or failed resampling population can reflect missing or degenerate heat and concentration information, the supplied starting values, the selected limit policy, the optimizer's interaction with the fitting surface, model complexity, or weak parameter identifiability. Resampling can fail even when the primary fit succeeds because each refit presents a different or reduced dataset to the same model.

CHAPTER 07

Multiple-experiment fitting

Multi-dataset fitting across processed Active experiments, including constraints, shared options, diagnostics, and combined results.

Last verified: 2026-09-23

Use **Multiple experiments** when several experiments should be analysed with the same model. Mark at least two processed experiments **Active** in **Analyze Data**. You can let each experiment have its own parameter values, or tell the fit that a supported parameter has a shared or temperature-dependent relationship across the experiments. This is useful when the experiments test the same interaction under different conditions.

When every constraint is **None**, the members are fitted independently. When one or more supported constraints are active, the solver performs one connected global optimization. Each experiment remains a member of the result, with member-specific parameters where the constraint state is **None**.

The model and model options apply across the active set. The resulting **Analysis Result** retains the member fits, any shared or temperature-dependent parameters, constraints, solver settings, diagnostics, and uncertainty output. Shared fitting controls are described in Single-experiment fitting.

Fit

The **Fit** tab identifies the **Mode** as **Multiple experiments** and exposes the shared **Model** selection. The solver controls include **Algorithm**, **Errors**, **Bootstrap**, **Limits**, **Weight by injection error**, **Concentration uncertainty**, and **Unlock parameters**. The available **Algorithm** values are **Nelder-Mead** and **Levenberg-Marquardt**; the error-estimation values are **None**, **Bootstrap residuals**, **Leave-one-out**, and **Profile likelihood**. For a globally fitted model, leave-one-out performs one deterministic refit per omitted experiment; profile likelihood uses the complete global objective and total observation/coordinate counts, while independent members use each member's local counts. Bootstrap count, concentration uncertainty, and parameter unlocking are disabled because they do not participate in deletion or profile schedules.

The **Result** controls describe whether a completed fit is stored as an **Analysis Result** and whether that result opens automatically. **Run Fit** and **Stop** are the fit controls. The status area records termination state, RMSD, iteration count, elapsed time, and error-estimation outcome when applicable. A multiple-experiment analysis is ready only when every member has usable processed data and the set contains at least two members.



A shared enthalpy constraint connects the three Active experiments in one global fit.

Parameters

The **Parameters** tab contains the exposed fit parameters and, in multiple-experiment mode, the **Global constraints** section. Parameter rows show the current value, units, uncertainty information when available, and the parameter lock state. Values and locks control individual parameters. Global constraints define how those parameters are related across experiments.

The constraint states have these meanings:

STATE	MEANING
None	The parameter remains member-specific. Each experiment has its own fitted value.
Same for all	One common value is fitted for every member in the set.
Temperature dependent	A supported parameter is represented across the temperature series by the relationship exposed by the model.
Thermodynamically linked	Affinity shares ΔG at the fixed fit reference and derives $\Delta G(T)$ from the selected enthalpy relationship, including ΔC_p when present.

The **Locked** state fixes an exposed parameter at its displayed value during fitting. **Locked** is not a constraint state and does not make a parameter common to the member experiments. A locked global value and a member-specific locked value therefore have different scopes.

For the core binding parameters, the available relationship states are model- and data-dependent:

PARAMETER	AVAILABLE STATES
Affinity	Independent, Shared Kd, Shared ΔG , or Thermodynamically linked
Enthalpy	None or Same for all; Temperature dependent is also available when the selected set exposes temperature dependence
N-value	None or Same for all
Offset	None or Same for all

The interface omits unsupported states for the current model. For affinity, **Independent** fits each member, **Shared Kd** fits one common dissociation constant, **Shared ΔG** shares one Gibbs-energy coordinate while Kd varies with temperature, and **Thermodynamically linked** derives the full temperature relationship from the selected enthalpy model. Profile linked intervals are approximate because fitted-coordinate correlations are omitted. Bootstrap and leave-one-out evaluate complete relationships, while Spolar–Record samples linked inputs independently in its Monte Carlo calculation.

For **Sequential Binding Sites**, the step count is one shared model option for the complete experiment set. The interface shows one **Affinity** constraint selector and one **Enthalpy** constraint selector, regardless of whether two, three, or four steps are active. A selected family style is applied to every active step while retaining a separate coordinate for each step: **Same for all** shares K_1 across members, separately shares K_2 , and so on; it does not force the different steps to equal one another. **Temperature dependent** likewise uses one ΔG_i per affinity step and one reference $\Delta H_i/\Delta C_{p,i}$ pair per enthalpy step. Reducing the step count discards inactive step values and constraints; increasing it later creates new defaults for those steps.

Offset is an energy-per-mole-of-injectant correction. For each injection, its absolute heat contribution scales with the injected amount—the injection volume multiplied by the syringe concentration. With **Offset: None**, each experiment has its own fitted molar offset. With **Offset: Same for all**, the **Parameters** tab exposes one common molar offset for the complete set. That shared value has its own **Locked** control; locking it fixes the common value for every member.

CALCULATION:

$$\Delta H(T) = \Delta H_{\text{ref}} + \Delta C_p(T - T_{\text{ref}})$$

$$\Delta G(T) = \frac{T}{T_{\text{ref}}} \Delta G_{\text{ref}} + \left(1 - \frac{T}{T_{\text{ref}}}\right) \Delta H_{\text{ref}} + \Delta C_p \left[\left(T - T_{\text{ref}}\right) - T \ln\left(\frac{T}{T_{\text{ref}}}\right) \right]$$

$$K_a(T) = \exp\left[-\frac{\Delta G}{RT} \right]$$

ΔH_{ref} is the enthalpy at the fixed fit reference, and ΔC_p is the equal-per-experiment regression slope when enthalpy is independent. The fit reference is fixed when the analysis is built; it is separate from the reporting temperature selected in preferences. A member's heat prediction continues to use its own fitted ΔH , while linked affinity uses the enthalpy trend. T and T_{ref} are absolute temperatures.

For independent enthalpies, the linked relationship gives every experiment equal weight, including repeated temperatures and locked enthalpies. Attaching profile results leaves the original best-fit values unchanged. Linked Gibbs and entropy intervals combine each fitted coordinate once, including its contribution through the enthalpy trend. Reported linked profile intervals are approximate propagation of the fitted coordinates and omit fitted-parameter covariance; bootstrap and leave-one-out results recompute the relationship from their participating members.

The image shows two side-by-side screenshots of the 'Parameters' tab in the FT-ITC Analysis software. Both screenshots have tabs for 'Fit', 'Parameters', 'Options', and 'Display', with 'Parameters' selected.

Left Screenshot (Before Locking):

- Global constraints:** N-value (None), Enthalpy (Same for all), Affinity (Temp. dependent).
- Enthalpy ΔH :** Value: -33.495 kJ/mol, Locked checkbox is ☐.
- Gibbs ΔG :** Value: -33.8053 kJ/mol, Locked checkbox is ☐.
- Fit section:** 'Run Fit' and 'Stop' buttons. Summary: 8 variables • 54 data points • 3 experiments. Will experiments fit globally.

Right Screenshot (After Locking):

- Global constraints:** Same as the left screenshot.
- Enthalpy ΔH :** Value: -33.495 kJ/mol, Locked checkbox is ☒.
- Gibbs ΔG :** Value: -33.8053 kJ/mol, Locked checkbox is ☐.
- Fit section:** 'Run Fit' and 'Stop' buttons. Summary: 7 variables • 54 data points • 3 experiments. Will experiments fit globally.

Global constraints define relationships between member parameters; Locked separately fixes an exposed value.

Options

The **Options** tab contains the options exposed by the selected model. In multiple-experiment mode, an option is a property of the shared model context and applies to every member. This includes model-specific concentration, stoichiometry, syringe-correction, or prebound-species settings when those options are exposed. The option values are preserved with the combined Analysis Result.

An option that changes which parameters are exposed can also change the available parameter rows and constraint states. For example, model options that share or replace stoichiometry alter which N-values are independently represented. The available rows always describe the active model and option combination.

Display

The **Display** tab controls the graph presentation for the active member set. Its controls cover the fitted line, residuals, error bars, confidence band, point labels, parameter box, excluded points, automatic scaling, unified axes, fitted offset, and fit-line interpolation. The parameter box can separately show model, fitted, and derived parameters.

Display settings affect the graph presentation, not the model, the member data, or the fitted values. The graph can therefore show the current member fits or connected global fit alongside member observations while the Parameters tab remains the source of the stored parameter and constraint state.

Combined Analysis Result

The combined **Analysis Result** contains the exact member experiments, the shared model and options, member parameters, any shared or temperature-dependent parameters, active constraints, solver and weighting settings, convergence diagnostics, and uncertainty results. Its result table can represent both global values and member-specific values. A selected member retains its own fitted curve and residual information even when a parameter is common across the set.

The status and diagnostics distinguish a successful solver termination from the quality and uncertainty of the fit. RMSD, iteration count, weighting, error-estimation method, and the number of successful uncertainty refits describe how the combined result was obtained. Wide or asymmetric uncertainty, a member with a markedly different residual pattern, or a member solution with an invalid status remains visible as member-level information rather than being hidden by a combined value. Result interpretation and validity are covered in Results and advanced analyses.

CHAPTER 08

Results and advanced analyses

Analysis Result views, validity, uncertainty presentation, evaluation temperature, and conditional advanced analyses.

Last verified: 2026-09-23

An **Analysis Result** saves a fit so you can inspect and export it later. Start with the fitted values and graph, then check where the predicted heats differ from the measured heats. The uncertainty display describes how precisely the parameters were estimated under the chosen model; the validity indicator tells you whether the saved fit still matches the experiment data currently in the project.

The result stores the model and options, constraints, solver state, weighting and uncertainty settings, member solutions, and a validity snapshot. The result workspace presents those stored values together with graph and analysis views. The fitting controls are described in Single-experiment fitting, and multi-dataset constraints are described in Multiple-experiment fitting.

Result views

The result view selector contains **Fit**, **Correlation**, and **Summary** for every Analysis Result. Thermodynamic temperature plots and parameter evaluation can show every active step in a sequential binding model. **Temperature** advanced analysis (Spolar Record method), **Salt**, and **Protonation** appear only when the selected model defines those analyses and the member metadata satisfy their additional requirements.

Summary presents the combined parameter graph and result table. The table can show fitted values, derived values, and the selected uncertainty representation for each stored solution.

Columns initially size to their headers and displayed values within compact limits; drag a header boundary to adjust an individual width. Wider tables retain their column widths and scroll horizontally, while spare space is assigned to the Experiment column. Molar-energy columns share one automatically resolved unit (or the fixed unit selected for result export), while ΔC_p columns resolve independently.

Selecting a row makes that member the current result solution; the selection is retained by the result workspace and drives **Fit** and the local portion of **Correlation**. When switching directly between analysis results, the selected experiment is retained when the destination contains a member with the same experiment ID. Use **Preferences > General > Energy units** to choose joules or calories.

Advanced analyses can sample the stored values and uncertainty summaries from profile-likelihood results. This does not provide a bootstrap ensemble or a joint model of parameter covariance. Saved leave-one-out refits can provide an envelope on the integrated-heats graph.

Fit presents the saved fitted curve, residuals, error bars, confidence band, and excluded points for the selected member. The graph is read-only: it represents the stored solution and does not expose fit controls or alter the underlying experiment.

Correlation presents a matrix calculated from residual-bootstrap refits. Its availability, scope, and interpretation are described under Parameter correlation.

Inspector tabs

The result inspector has four tabs with shared labels across the supported desktop versions:

Summary, Analysis, Experiments, and Model.

The **Summary** tab contains the result identity, model, member count, RMSD, validity, information criteria, and solver diagnostics. Solver information includes algorithm, iterations, whether injection-error weighting was used, error-estimation method, and bootstrap count.

In the browser viewer, RMSD is shown as a saved unweighted display diagnostic in μJ , separate from the weighted fitting objective. When the saved convergence record contains it, **Molar RMSD (kJ/mol)** is shown separately: the result summary uses the saved global metric, while an individual fit uses that fit's saved metric. Missing or non-finite saved values remain unavailable; member values are not averaged to reconstruct the result metric.

Result validity

Validity answers whether the saved fit still describes the experiment inputs currently in the project. FT-ITC compares the result's saved inputs with the current member experiments, including their concentrations, cell volume, injection bookkeeping, included injections and heats, relevant processing and attributes, and tandem segment layout. Changing a result name or attaching a different solution to an experiment does not by itself change this status. Validity does not say whether the model is scientifically appropriate or the fit is good.

- **Analysis is valid** means the saved fit inputs still match.
- **Partially invalid** means some members of a multiple-experiment result fitted independently have changed, while others still match. A shared-parameter fit is treated as invalid when any member changes.
- **Invalid** means the saved fit no longer matches the current inputs, or the set of member experiments has changed.
- **Unknown status** means FT-ITC cannot make the comparison, for example because an older result has no validity snapshot.

The validity section gives reasons for a mismatch. Repeated desktop reasons affecting more than two experiments are summarized by category and count. If inputs changed, inspect those reasons and use **Update Result** to fit the current data. Until then, the stored parameters still describe the earlier inputs. In the browser viewer, **Saved result has analysis warnings** is separate from validity: it flags issues such as a fit reaching a parameter boundary or an optimizer limit, even when the saved inputs still match.

Information criteria

AIC and **AICc** help compare candidate models fitted to the same observations. They balance agreement with the data against the number of fitted parameters; a smaller value is preferred within a comparable set. They do not tell you that any model is scientifically correct.

The **Information criteria** section reports the analysis-level AICc when it is available, otherwise AIC, together with the included observation count n and likelihood parameter count K . Both weighted and unweighted criteria estimate one residual-variance parameter, so $K = p + 1$, where p is the number of free fitted parameters. Weighted criteria use the injection integration errors as relative uncertainties and estimate one common variance multiplier from the standardized residuals. AICc is unavailable when $n \leq K + 1$, in which case AIC is shown instead. If the likelihood cannot be evaluated, the displayed criterion shows its diagnostic reason.

When members were fitted independently, the result table also includes an **AICc / AIC** column. It reports each member's own criterion and is intended for comparing alternative models fitted to the same experiment, observations, response definition, and weighting mode. Values in different experiment rows are not comparable because they use different observations. AICc is shown when available, with AIC as the fallback when the small-sample correction is undefined;

Unavailable means that the member likelihood could not be evaluated. Shared-parameter global fits expose only the analysis-level criterion and omit this member column. Neither AIC nor AICc establishes model adequacy.

The criteria use the saved result's included injections and response definition. At analysis level, residual statistics are pooled before estimating the variance, including when members were fitted independently. Unweighted criteria estimate one common variance across all members; weighted criteria estimate one common multiplier of the injection-error variances. Each independent member's own criterion estimates its variance separately. Consequently, neither pooled AIC nor pooled AICc is a sum of the member values.

With residuals r_i , raw residual sum of squares $RSS = \sum r_i^2$, integration-error SDs σ_i , and standardized residual sum of squares $Q = \sum (r_i/\sigma_i)^2$, the maximized likelihood terms are:

CALCULATION:

$$-2\log L = n \left[\log \left(\frac{2\pi RSS}{n} \right) + 1 \right]$$

This estimates one shared residual variance from the total squared mismatch RSS, then scores how likely the observed heats are under that estimate.

For a **weighted fit with one estimated variance multiplier**:

CALCULATION:

$$-2\log L = n \left[\log \left(\frac{2\pi Q}{n} \right) + 1 \right] + \sum \log \sigma_i^2$$

This uses each injection's processing SD as a relative weight. The score also includes their different uncertainty scales.

The weighted variance multiplier is Q/n , calculated analytically without adding an optimizer variable or changing the stored integration errors or fitted parameters. New weighted fits require a finite, positive SD for every included injection. Zero residual variance makes the estimated-variance likelihood unavailable. Weighted profile-likelihood intervals use the unchanged processing SDs as relative weights and apply the same overall residual-scale interpretation with an F-calibrated threshold. Neither calculation writes a model-derived value back to an injection SD.

The fitted parameter count p includes only parameters free in the saved global model. Shared coordinates count once; member-specific coordinates count once per member. For a member criterion, p includes only that member model's free fitted parameters. In both weighting modes, the reported values use $K = p + 1$, $AIC = -2 \log L + 2K$, and $AICc = AIC + 2K(K + 1)/(n - K - 1)$. This standard small-sample correction is an approximation for nonlinear ITC models.

Smaller values are preferred only when comparing models that use the same observations, response definition, and weighting mode. AIC and AICc do not establish model adequacy or replace residual and scientific checks. Prefer AICc when it is available.

Information criteria are recalculated when a saved project is opened. Weighted values from the former fixed-sigma calculation can therefore change; compare results calculated under the same likelihood convention.

Summary combines result validity, stored solver information, display controls, and the member parameter overview.

The **Analysis** tab contains the result view selector, parameter evaluation, and the analysis-specific controls and outputs. It is also the location of the uncertainty display, correlation information, and evaluation-temperature presentation associated with the selected view. Energy units are chosen consistently from the displayed central values and are also used for graph axes, errors, fitted bands, tooltips, and parameter lists.

The **Experiments** tab lists the result members and their stored status and condition information, including member temperature. The row selected in the result table determines which experiment appears in **Fit**; this tab provides that experiment's details.

The **Model** tab shows the stored model options, locked parameters and their fixed values, and the active constraints. A constraint with state **None** is not listed as an active global constraint; **Same for all** and **Temperature dependent** entries identify the relationships retained by the Analysis Result. Affinity constraints are labelled **Independent**, **Shared Kd**, **Shared ΔG**, or **Thermodynamically linked** to describe the fitted relationship precisely; enthalpy and other parameters retain their own constraint labels.

Uncertainty and evaluation temperature

The central value is the model's best fit to the original data. An uncertainty display gives a range or spread around that value under the selected estimation method. It does not include every possible experimental error or guarantee that the chosen model is correct.

The **Errors** display control provides **Automatic**, **Standard deviation**, **95% confidence interval**, and **SD + 95% CI**. **Standard deviation** presents the primary best-fit value with a symmetric $\pm$ SD; **95% confidence interval** presents that same best-fit value with its lower and upper confidence limits; and **SD + 95% CI** presents both. The central value is always the primary best fit, not the mean or median of the resampled values. Residual-bootstrap intervals use percentile limits; profile-likelihood intervals use the likelihood-threshold endpoints, with an equivalent symmetric scale for the SD display.

Automatic makes this choice separately for each reported quantity. Let L and U be its stored 95% confidence limits and $\hat{\theta}$ its primary best-fit value:

CALCULATION:

$$w_{\text{lower}} = \hat{\theta} - L; \quad w_{\text{upper}} = U - \hat{\theta}$$

$$\text{asymmetry} = \frac{|w_{\text{upper}} - w_{\text{lower}}|}{w_{\text{upper}} + w_{\text{lower}}}$$

Automatic shows the 95% confidence interval when both widths are positive and the asymmetry is at least 0.18; otherwise it shows SD. This is a display rule based on interval imbalance around the best fit, not a formal statistical test of distribution skewness.

The decision is made after a fitted coordinate has been transformed into the displayed quantity. A nonlinear transformation—such as fitting $\log_{10}(K_a)$ and displaying K_d —can therefore make the displayed interval asymmetric and cause **Automatic** to select CI for that quantity.

Changing **Errors** changes only how stored uncertainty is presented in tables, parameter evaluation, and graphs. It does not rerun the fit, change the best-fit parameters, or turn one error-estimation method into another. Bootstrap construction, parameter transformation, and the SD and percentile calculations are described under Parameter uncertainty.

The **Parameter Evaluation** section contains an evaluation **Temperature** field and the displayed thermodynamic quantities at that temperature. Temperature display can be **Celsius** or **Kelvin**. Changing the evaluation temperature updates the displayed quantities calculated from the stored model; it does not change injection heats or refit the result. Temperature-dependent values are meaningful together with their model, units, uncertainty representation, and evaluation temperature.

For locally fitted member parameters, summary values include individual parameter uncertainty. **Combined SD** is the square root of the between-experiment sample variance plus the mean individual variance; temperature trends use residual variance in place of sample variance. It describes combined spread, not the standard error of the mean, and observed spread may already contain fitting noise. **Approximate propagated interval** targets the average or evaluated trend: individual lower and upper 95% interval widths are propagated separately and combined with the observed-spread contribution. The best-fit central values remain unchanged, and asymmetric intervals can remain asymmetric. Exactly two observations at distinct temperatures propagate individual errors without estimating residual variance; additional observations, including replicates at those same temperatures, contribute residual spread. ΔC_p propagates the corresponding slope uncertainty. These intervals do not have established 95% coverage, especially with few experiments, and covariance between experiments is omitted even for local parameters coupled through a global fit. Shared and temperature-constrained parameters retain their model-estimated uncertainty and CI95 meaning.

Parameter correlation

This view shows whether two fitted parameters tend to move together when the data are resampled. A strong relationship can mean that the data have difficulty separating their effects, but the display alone cannot establish why.

Correlation shows Pearson correlations between fitted parameter coordinates across residual-bootstrap refits. It requires **Bootstrap residuals**, at least 30 complete refits, and at least two parameters that vary across those refits. Parameters with no variation are omitted. Parameters fixed in the primary fit are also omitted unless **Unlock parameters** allowed them to vary during bootstrap error estimation; such parameters are marked with an asterisk and a warning that bootstrap parameter unlocking was enabled.

CALCULATION:

$$r_{jk} = \frac{\sum_b [(\theta_{bj} - \bar{\theta}_j)(\theta_{bk} - \bar{\theta}_k)]}{\sqrt{\left[\sum_b (\theta_{bj} - \bar{\theta}_j)^2 \sum_b (\theta_{bk} - \bar{\theta}_k)^2 \right]}}$$

r_{jk} is the Pearson correlation between fitted coordinates j and k . The index b runs over complete residual-bootstrap refits, and $\bar{\theta}$ is the corresponding mean coordinate.

For each off-diagonal cell, the application also reports an approximate 95% Monte Carlo precision interval using Fisher's transformation:

CALCULATION:

$$z = \operatorname{atanh}(r); \quad z_{\pm} = z \pm \frac{1.959964}{\sqrt{B-3}}; \quad r_{\pm} = \tanh(z_{\pm})$$

B is the number of complete refits: usable refits with finite values for every displayed coordinate. The diagonal is the structural self-correlation $r = 1$ and has no Fisher precision interval.

Matrix values range from -1 to $+1$; pointing to a cell shows r and the Monte Carlo precision interval; an interval spanning zero marks the sign as unresolved at the current simulation precision. The result panel separately reports attempted, usable, failed, and complete refits. It warns when fewer than 100 complete refits make precision coarse, at least 20% of attempted refits failed, a usable refit lacked a finite displayed coordinate, a sign is unresolved, or the matrix is structurally rank limited. The correlation matrix remains unavailable below 30 complete refits. For a single-experiment result, the scope is **Single experiment**. A multiple-experiment result can show **Shared** coordinates alone or **Shared + selected local** coordinates when a member is selected. Shared and local labels identify the scope of each parameter.

Sequential correlations use the actual fitted coordinates. Affinity coordinates are labeled **log10 Ka1** through **log10 Ka4**, active enthalpy coordinates are included, and no N-value coordinate is added. A global result shows the shared per-step coordinates and, when requested, unconstrained coordinates for the selected member without duplicating constrained member values.

INTERPRETATION:

Correlation shows how fitted coordinates varied together under the residual bootstrap. The Fisher interval describes the finite-bootstrap Monte Carlo precision of the correlation coefficient r itself; it is not a confidence interval for either fitted parameter and does not replace the parameter uncertainty display. It does not establish identifiability, causality, model adequacy, or model validity. An interval spanning zero means only that the sign is unresolved at this Monte Carlo precision. Frequent refit failures can make the retained ensemble selective, and the Fisher interval does not account for those failures. Affinity is evaluated in the fitted coordinate system— $\log_{10}(K_a)$ —rather than as the displayed K_d . A rank warning states the structural limit $\text{rank} \leq B - 1$ for the displayed parameter count; it is distinct from numerical rank and scientific model validity.

Advanced analysis views

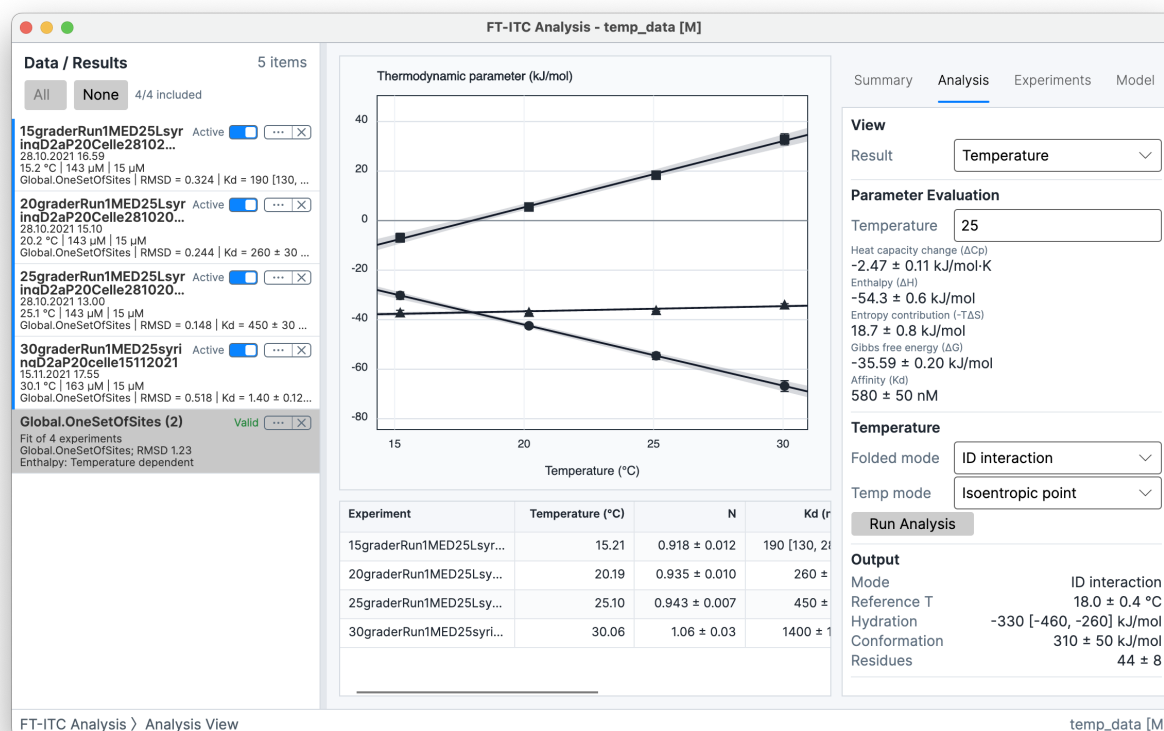
These views ask how a fitted interaction changes with experimental conditions. **Temperature** examines a series measured at different temperatures, **Salt** examines a series with different salt conditions, and **Protonation** compares experiments in different buffers. Their conclusions depend on the recorded conditions and the underlying fit.

All advanced analyses require a **One-Set-Of-Sites** Analysis Result. Each analysis also requires the relevant variation in experimental conditions and the corresponding metadata. The advanced analyses operate on the stored member solutions and expose their own calculated outputs; they do not change the base fit parameters. A sequential result can still show its ordinary per-step ΔH , ΔG , $-T\Delta S$, K_d , and temperature-dependence presentation; it reports the Spolar Record method, protonation, and electrostatics as unsupported by that model rather than hiding the ordinary thermodynamic views.

Temperature

The **Temperature** view is available when the difference between the highest and lowest member temperatures exceeds the configured minimum span. Its temperature-analysis controls expose **Folded mode** values **Globular** and **ID interaction**, together with **Temp mode** values **Isoentropic point**, **Mean temperature**, and **Reference temperature**. For a thermodynamically linked result, the isoentropic temperature is derived from the complete Gibbs, enthalpy, and heat-capacity relationship. Profile intervals are propagated approximately and omit fitted-coordinate correlations; bootstrap and leave-one-out uncertainty evaluates each saved relationship jointly, while the Spolar–Record Monte Carlo samples its linked inputs independently.

The stored temperature-analysis output includes reference temperature, hydration contribution, conformational contribution, and residue estimate. These values describe the selected folded and temperature-evaluation modes under the fitted temperature dependence. They remain conditional estimates of the stored model and member series rather than direct structural measurements.



Temperature presents the stored temperature dependence together with its evaluation and folded-state analysis controls.

Salt

The **Salt** view requires a **Salt** attribute for every member and sufficient ionic-strength span across the member set. Its **Graph mode** values are **Affinity vs Salt**, **Debye-Huckel**, and **Counter Ion Release**. Counter-ion release is a salt analysis mode and is presented in the Salt view rather than as a separate advanced analysis.

The Salt output includes the extrapolated **Kd0** and **Counter ion** result when the analysis has a calculated fit. The graph mode determines whether the displayed dependence is expressed against salt, ionic strength using Debye-Huckel behavior, or ion activity for counter-ion release. The result is limited by the recorded salt identities, ionic-strength values, and the quality and span of the member affinity values.

CALCULATION:

$$\ln K_d(I) = \ln K_{d,0} + s\sqrt{I}$$

$$\ln K_d = b + n_{\text{ion}} \ln a_{\text{ion}}$$

I is ionic strength, s is the fitted sensitivity, a_{ion} is ion activity, and b is the fitted intercept. $K_{d,0}$ is the extrapolated value in the Debye–Hückel view. n_{ion} is the reported slope for Counter Ion Release; its sign is reported by the analysis and is not assigned an interpretation here.

Protonation

The **Protonation** view requires a **Buffer** attribute for every member and at least two distinct buffer identities. Its calculated output contains **Protons** and **Binding H**, with uncertainty when the stored analysis contains the corresponding uncertainty results.

The view relates the stored member binding enthalpies to the buffer protonation information associated with their conditions. The result is conditional on the recorded buffer identities, their temperature-dependent protonation enthalpies, and the model assumptions; it does not identify a particular residue or microscopic protonation event.

CALCULATION:

$$\Delta H_{\text{obs}} = \Delta H_{\text{bind}} + m \Delta H_{\text{buffer}}$$

The fitted slope is m . The application reports **Protons** as $-m$, while **Binding H** is the fitted intercept at zero buffer protonation enthalpy. This sign convention follows the application's protonation-enthalpy convention; it does not by itself assign a microscopic uptake or release mechanism.

Advanced-analysis values are supplemental views of a stored Analysis Result. Their availability and outputs are determined by the One-Set-Of-Sites model, member variation, metadata, selected graph or evaluation mode, and any completed uncertainty calculation. Result validity remains a separate indication of whether the stored fit inputs match the current project state. Figure and table output is covered in Figures and export.

The buffer registry contains approximations and some unknown quantities, so confirm that the chosen buffer values apply to the experiment. The TAPSO pKa temperature correction is a local approximation for 20–30 °C. Its protonated zwitterion is neutral, and imidazole's ionization heat-capacity slope is $-9 \text{ J}/(\text{mol K})$. Recalculate affected derived analyses when comparing with results produced before these corrections.

CHAPTER 09

Figures and export

Configure final and supporting figures, print active graphs, and export data, results, figures, and complete analysis reports.

Last verified: 2026-09-23

Choose a figure when you want to show a graph, a data export when you need injection values, and a result table or report when you need to share fitted quantities and their context. Check the selected experiments, units, and uncertainty display before sending an output.

FT-ITC Analysis keeps publication figures, numerical data, and fitted result tables as separate output types. **Final Figure** is an Experiment Data workflow. An Analysis Result does not open directly as a Final Figure; **Export Associated Final Figures...** on a result exports figures for the experiment solutions associated with that result.

Final Figure

The **Final Figure** workspace renders the selected Experiment Data as a publication figure. Its preview and PDF output reflect the selected experiment, processing state, fitted solution, and figure options. To export figures using the fits stored in an Analysis Result, use **Export Associated Final Figures...** on that result.

The **Export PDF** controls offer three scopes:

- **Selected** opens a save dialog for the displayed experiment figure and suggests the experiment name as the PDF filename.
- **Active** exports figures for Active Experiment Data. **All** exports figures for all Experiment Data in the project.

For **Active** and **All**, choose a parent folder and the app creates or reuses a subfolder named after the saved project file (without its extension). For an unsaved session, it uses the local export date (`yyyyMMdd`) as the folder name. Existing PDFs are kept; when a filename is already in use, the new figure gets a numeric suffix such as `Experiment (2).pdf` .

The result-list command **Export Associated Final Figures...** writes one final-figure PDF per member experiment using that result's saved fit. Each saved solution is attached only while its figure is made; the experiment's previous solution is restored afterward. It is available for a result with exportable solutions. See Results and advanced analyses for result validity and stored member-solution behavior.

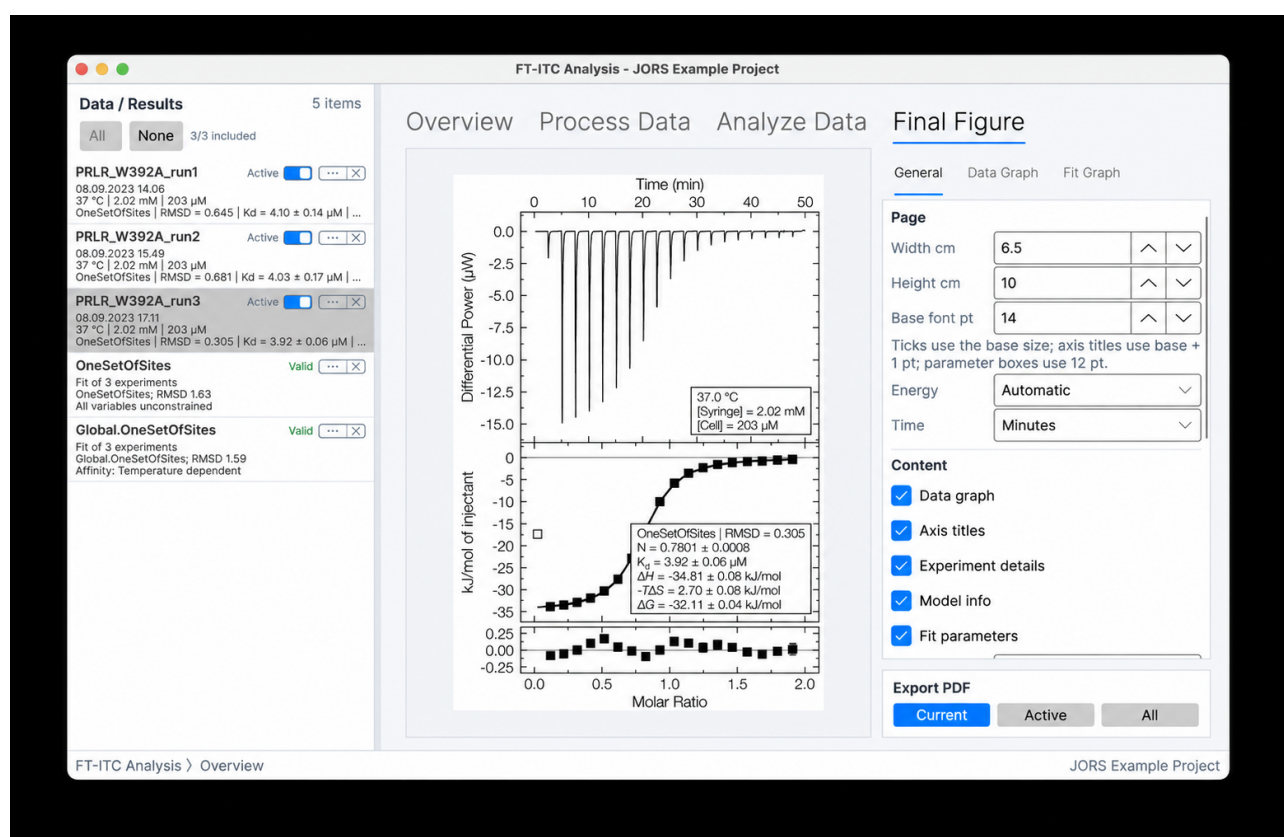
WHERE FINAL FIGURE SETTINGS ARE USED

- **Final Figure PDFs**, including Active, All, and **Export Associated Final Figures...**, use the current Final Figure appearance controls.
- **Supporting Figure...** starts with a snapshot of those controls when its window opens. This includes units, axis limits, visible traces and fits, and excluded-point automatic Y scaling. Its own plot size, grid, typography, panel labels, grouping, and information-box controls take precedence; later Final Figure edits do not update an already open Supporting Figure.
- **Analysis Report** uses a separate compact figure preset and its own energy and uncertainty selections. Final Figure axis limits and excluded-point automatic Y scaling do not carry over to report figures.

General

The **General** tab defines the page and common content. Page controls specify width and height in centimeters and the base font size. The **Energy units** selector provides **Automatic**, **J**, **kJ**, **cal**, and **kcal**. **Automatic** chooses a common molar-energy unit based on the figure's plotted and reported central values. A fixed choice uses the selected energy unit throughout the figure, except for the thermogram units described below. The selected family also controls thermogram units: differential power is **μW** or **μcal/s**, while integrated heat is **μJ** or **μcal**. Time controls set the displayed time unit. Content controls include the data graph, axis titles, experiment details, model information, fit parameters, and the information-box placement. The uncertainty selector provides **Automatic**, **SD**, **CI**, **SD + CI**, and **None**. **Automatic** uses the same per-quantity asymmetry rule described under Uncertainty and evaluation temperature.

The parameter controls determine which information appears in the information box: thermodynamic, derived, or offset parameters; temperature; concentrations; injection delay; instrument; and user-defined attributes. The information box is descriptive figure content and does not alter the underlying fit.



Final Figure combines a live publication preview with page, content, and PDF export controls.

Data Graph

The **Data Graph** tab controls the differential-power trace. Power and time axis titles, tick density, and explicit minimum and maximum values define the axes. **Corrected data** selects the baseline-corrected trace when one is available. **Shared power axis** applies one power-axis range across the active experiments represented by a figure set.

Baseline controls expose the baseline, its **Solid** or **Dashed** style, **Under data** or **Over data** layer, and line width. **Integration ranges** displays the integration intervals as **Bar**, **Fill**, or **Endpoint lines**. These overlays describe processing and do not recalculate integration.

Fit Graph

The **Fit Graph** tab controls the integrated-heats and model panels. Enthalpy and molar-ratio axis titles, tick density, and explicit ranges define the axes. Symbols are **Square** or **Circle**, with an independent point size. **Shared X axis** and **Shared enthalpy axis** unify the corresponding ranges across active experiments; the residual-axis range follows the shared enthalpy setting when configured for unified residual axes.

Include excluded points in automatic Y scaling lets excluded injections affect the automatic integrated-heat and residual axis ranges. It starts from the application preference. Explicit axis limits take precedence.

Fit line displays the fitted binding curve with a selected width and **Smooth**, **Spline**, or **Linear** smoothness. **Show residuals graph** adds the residual panel, and **Residual gap** separates it visually from the fit panel. The display controls include the zero enthalpy line, confidence band, error bars, excluded points, excluded error bars, and offset-corrected heats. Error bars use processed injection uncertainties; confidence bands require bootstrap uncertainty in the fitted solution; fit lines and residuals require a fitted solution.

Data Graph and Fit Graph provide separate controls for the thermogram and fitted-heats panels.

Numerical data export

File > Export Data... opens the numerical export dialog. **Export Selected Data...** invokes the same export for the selected experiment. The data scope is one of **Selected experiment**, **Active experiments**, or **All experiments**.

The format list contains:

- **Thermogram Data** — a `.csv` file containing time and power samples, with the **Export baseline-corrected trace** option when baseline-corrected samples exist.
- **Integrated Peaks** — a `.csv` file containing injection-axis values, molar heats, their SD, fitted values, and residuals when available, with **Export offset-corrected peaks** when a fitted solution is available.

- **Combined Data** — a `.csv` file placing thermogram samples and integrated-peak columns side by side; raw, corrected, fitted, and residual columns are included when available.
- **MicroCal / SEDPHAT** — a `.dat` file with MicroCal-style `DH`, `INJV`, `Xt`, `Mt`, `XMt`, `NDH`, `DY`, and `Fit` columns. `DH` is in μcal , `INJV` is in μL , concentrations are in mM, and `NDH`, `DY`, and `Fit` are in cal/mol. The first injection uses `--` for normalized heat and residual, following the conventional excluded-first-injection layout.
- **pytc** — a `.dh` file containing the pytc-compatible injection and metadata fields.
- **ITCsim** — a `.csv` file containing ITCsim-compatible injection data and metadata, with offset-corrected peaks available when a fitted solution exists.

Molar-energy unit is available for Integrated Peaks, Combined Data, and ITCsim: choose **J/mol**, **kJ/mol**, **cal/mol**, or **kcal/mol**. Each export dialog starts at kJ/mol for the Joules energy preference or kcal/mol for the Calories preference; a choice applies to that export only. Integrated Peaks and Combined Data identify the chosen unit in the heat, SD, model, and residual column headers. ITCsim keeps its `PeakHeat` column name and records the unit as `#EXPINFO ENERGYUNIT J`, `KJ`, `CAL`, or `KCAL`; those tokens denote energy per mole. Peak area is an absolute heat in J and is distinct from the exported molar heat.

Other output units remain format-specific. Thermogram samples use seconds and watts; MicroCal/SEDPHAT and pytc use their documented concentration, volume, temperature, and heat conventions. Fitted columns and correction controls are disabled when the selected data do not contain the corresponding processed or fitted state. Export defaults are described in Settings and defaults.

These exports are not project backups. General `.csv` and `.tsv` exports cannot be reopened through **File > Open...** A MicroCal / SEDPHAT `.dat` export or pytc `.dh` export can be reopened as integrated-heat input, but doing so restores neither a thermogram nor the FT-ITC project state and requires choosing the source heat unit. Select **microcalorie** for the `DH` column when reopening a newly generated MicroCal / SEDPHAT export. Historical FT-ITC `.dat` exports may instead require **joule**, matching the unit used when they were created. Save an `.ftxtc` project when the analysis must remain editable.

Analysis Result Exporter

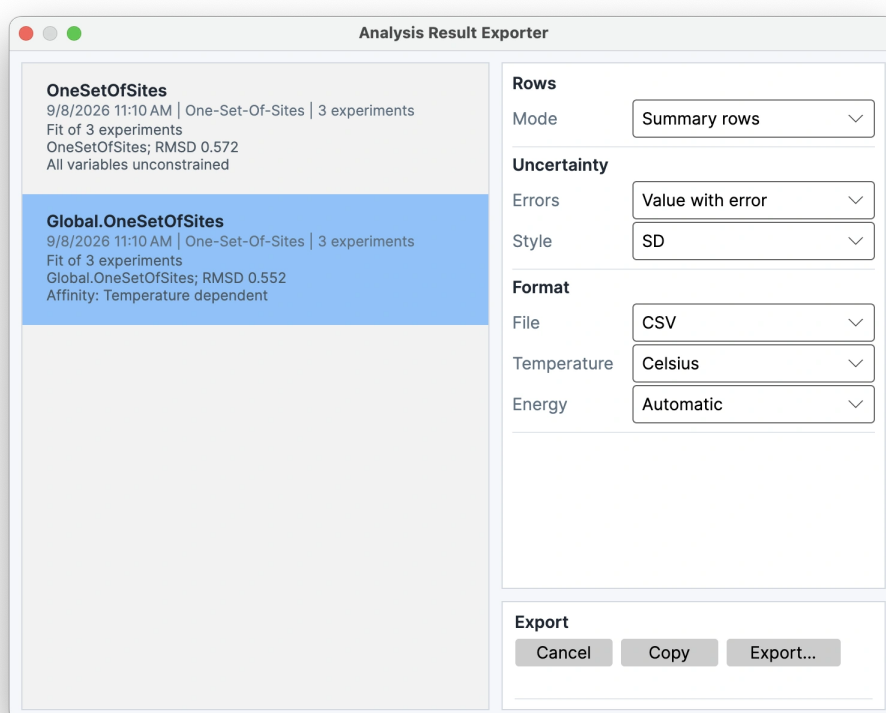
Analysis Result Exporter... builds a table from one or more selected Analysis Results. **Summary rows** produces one row per result; **All replicate rows** produces one row per member fit. Error layout is **Value with error** or **Separate columns**. Uncertainty style is **SD**, **CI**, or **SD + CI**. The file format is **CSV** or **TSV**. **Energy units** provides **Automatic**, **J**, **kJ**, **cal**, and **kcal**. Automatic chooses one stable molar-energy unit across all selected results and one independent unit for ΔC_p columns; a fixed choice uses the selected energy unit consistently. Temperature presentation is **Celsius** or **Kelvin**.

The configured table is available through **Copy** and **Export...** Copy places the same delimited text on the clipboard; Export writes it to a file. Profile-likelihood SD columns use the equivalent display scale implied by their stored endpoints and are labeled as such in exported metadata; the exact lower/upper endpoints remain available in separate-column mode. The exporter does not include the parameter-correlation matrix, which is calculated for display rather than stored as a result-table field.

Summary rows use the same Parameter Evaluation calculation at its default evaluation temperature: the configured reference temperature for a temperature series, otherwise the mean experiment temperature. The exported temperature column labels that evaluation temperature. Local summary rows report **Combined SD** and an **Approximate propagated interval**, using the same calculation described under Parameter Evaluation. Summary interval columns have neutral `_interval_lower` and `_interval_upper` suffixes; individual rows retain their CI95 columns. Kd is derived from the summarized Gibbs energy rather than averaged directly, and ΔC_p columns are included for applicable temperature-dependent results. No results are pooled across separate analyses.

Result tables, clipboard output, final-figure metadata, and viewer output create thermodynamic columns from the fitted model shape. A sequential result therefore exports one Kd, ΔH , ΔG , and $-T\Delta S$ value for every active step, including steps 3 and 4, rather than truncating the result to two interactions. The fixed step count is included as model information. Values remain in their displayed families; exporting does not convert macroscopic sequential constants into microscopic site constants.

Each affinity column chooses its concentration unit independently from its own magnitude. For example, Kd1 may be shown in nM while Kd4 is shown in μM . The column header and its values always use the same unit; this is display scaling only and does not change fitted or persisted values.



Analysis Result Exporter configures a delimited table for clipboard or file output.

Analysis Report

Tools > Analysis Report... creates a complete, printable report from one or more saved Analysis Results. The same tool is available as **Export Analysis Report...** in result-specific menus, where that result is initially selected. Use **Select report contents...** to choose results and optional supporting experiments together. The picker follows application order; applying the same ordered selection again restores its saved study context and approved interpretation. At least one result is required.

Analysis Report

Interpretation Preview

Zoom 100%

FT-ITC Analysis 1.5.0 OneSetOfSites Page 1 of 19

FT-ITC Analysis Exported 17 Sep 2026 UTC

Interpretation

Overall interpretation

Result 1 supports a reproducible, predominantly single-transition binding isotherm across 15–30°C. The one-set-of-sites fits give apparent affinities in the submicromolar range and *N*-values close to one. The temperature series shows a coherent trend: affinity weakens as temperature increases, while binding enthalpy becomes more exothermic.

Experiment	K _d (μM)	Δ <i>H</i> (kJ/mol)	<i>N</i> ; RMSD (kJ/mol)
Experiment 1A	0.163	-29.6	0.917; 1.09
Experiment 1B	0.339	-44.4	0.935; 0.698
Experiment 1C	0.469	-55.0	0.943; 0.519
Experiment 1D	0.771	-58.4	1.027; 1.20

Across the series, K_d increases approximately 4.7-fold from 15 to 30°C. Δ*G* remains favorable but becomes less favorable by about 2.0 kJ/mol. The fitted temperature slopes are clearly non-zero within the reported fit-based intervals: Δ*H* changes by -1.97 ± approximately 0.10 kJ/molK, the entropy-contribution coordinate changes by +2.09 ± approximately 0.14 kJ/molK, and Δ*G* changes by +0.125 ± approximately 0.040 kJ/molK. Thus, the temperature dependence is supported by the fitted data, although its molecular interpretation is not established by these measurements alone.

Experiment observations

The fit quality is strongest at 20–25°C. Experiment 1B and Experiment 1C have RMSDs of 0.698 and 0.519 kJ/mol, respectively, with no standardized residual above two injection errors. Experiment 1A has the largest fitted offset, -10.33 kJ/mol, the largest baseline drift, 1.80 μW, a residual slope of -355 J/mol per molar-ratio unit, and one standardized residual above two errors. This makes the 15°C endpoint less secure than the central temperatures.

Experiment 1D also deserves attention: its RMSD is 1.20 kJ/mol, injection 8 at molar ratio 0.800 was excluded, and the corresponding stored residual before exclusion is -8.36 kJ/mol. Because this injection lies within the transition region, its exclusion could affect the 30°C affinity, enthalpy, *N*-value, and the temperature slopes. The exclusion appears associated with an anomaly rather than an ordinary observation, but the reason is not documented.

N increases systematically from 0.917 at 15°C to 1.027 at 30°C. The 30°C interval includes one, whereas the lower-temperature intervals remain below one. This is a modest but structured pattern, not evidence by itself for a non-integer molecular stoichiometry. It is compatible with temperature-dependent effective concentration or activity, partial inactivity, preparation differences, or limitations of the single-site representation.

Uncertainty estimation completed successfully: all 100 residual-bootstrap replicates were successful for each member fit, and all 400 result-level uncertainty fits succeeded. However, the bootstrap correlations are substantial, especially between Δ*H* and log10 affinity and between Δ*H* and offset. The parameters should therefore not be interpreted as independent measurements.

Build Analysis Report

Choose the result and content to include in the PDF.

Report contents

1 result
One-Set-Of-Sites • 4 experiments • Valid • 0 supporting experiments

Document

Title OneSetOfSites
Subtitle

Presentation

Energy Joule
Temperature Celsius
Uncertainties SD + 95% CI

Optional content

☒ Injection tables
☒ Condense repeated experiments
☒ Temperature dependence
☒ Parameter correlations

Select all Clear

Interpretation

PDF export

19 A4 pages ready.

Cancel Update Preview Export...

Analysis Report combines a paginated preview with controls for report contents, presentation, interpretation, and PDF export.

Report contents and layout

The report is a multipage A4 portrait PDF with margins suitable for printing. A report containing multiple results begins with a combined result index and concise table of contents on the first page, followed by the report-level interpretation and one self-contained chapter for each result. Results are referenced as **1**, **2**, and so on, and their experiments as **1A**, **1B**, **2A**, **2B**, and so on. These labels are derived from the saved result and experiment order, so figures, captions, and interpretation evidence use the same stable references. No experiments or scientific estimates are merged across results. Each chapter overview arranges the experiments in compact, borderless figure panels labeled with their references and names. The report automatically chooses three to five columns, centers incomplete rows, and suppresses repeated axis titles.

The analysis summary contains a compact grouped thermodynamic bar chart for every saved member and active binding step. Its whiskers show only the saved 95% confidence interval from the solution distribution or profile-likelihood calculation; the symmetric SD approximation remains available in report tables but is not superimposed on this plot. Each experiment section places a wider, baseline-focused view of the uncorrected thermogram—with its saved baseline and compact horizontal markers for the integration windows—beside a top-aligned Final Figure at its standard size containing the complete thermogram, followed closely by fitted and derived parameters. **Fit details** also reports dimensionless c-values where the fitted model defines them. The one- and two-site Wiseman values use $c = N[\text{cell}]_0/K_d$; syringe-correction fits use the fixed site count rather than the fitted syringe active fraction. Sequential fits report $[\text{cell}]_0/K_d$ for each active step, while competitive fits report the apparent c-value based on K_{d_app} . When **Injection tables** is selected, a table records every saved injection, its inclusion state, volume, concentrations, analysis-axis value, integrated heat and uncertainty, fitted heat, and residual. This processing presentation is an automatic report preset and does not add another figure control. A concise notice replaces the processing graph when raw thermogram samples are unavailable. Selected shared correlations follow the analysis summary; selected member correlations follow that experiment's parameter table, using a compact native matrix rather than a dedicated page.

Directly selected experiments that are not represented by a selected result appear once in a report-level **Supporting experiments** chapter after the result chapters and are labeled **S1**, **S2**, and so on. The report uses only saved content: a raw thermogram with any saved baseline and integration boundaries, finite integrated heats explicitly labeled as having no fit, experiment metadata, notes about processing or correction exceptions, dates with a verified source, attributes, comments, and the report-wide optional injection table. Missing stages receive concise availability notices. Experiments already represented by a checked result remain visible but unavailable as supporting selections. Recorded buffer-subtraction references may be selected automatically according to the application setting, but remain ordinary optional selections; the report does not judge whether a reference is an appropriate blank. Report creation never fits, reintegrates, or performs subtraction.

Configure the report

Choose a title, an optional subtitle, energy and temperature units, and an **Uncertainties** style. The subtitle appears as secondary, regular-weight text below the report title. Reports default to **SD + 95% CI**. The selected style controls uncertainty in tables and applicable analysis plots. The thermodynamic summary bar chart always shows available saved 95% confidence intervals, as described above; it does not add SD whiskers. This selection is remembered only for the current application session. Final Figure injection errors and fitted confidence bands retain their established scientific meanings.

The subtitle editor accepts longer descriptions. It wraps text and scrolls vertically so you can review it before previewing the report.

Optional content includes report-wide **Injection tables** and **Condense repeated experiments** switches and the completed analysis types available in any selected result. Each option applies to every eligible result chapter; chapters without that saved content simply omit it. The injection-table switch is on by default and applies consistently to every experiment. When the same experiment occurs in a later result, condensed mode retains both figures, comments, source file, date with a verified source, instrument, temperature, concentrations, and all result-specific fitted content, but replaces repeated secondary acquisition and processing details with a reference to

its first occurrence. The **Parameter correlations** option includes the shared matrix and all available member matrices alongside their corresponding summaries. Available analyses are selected by default; **Select all** and **Clear** change that analysis list together. Temperature dependence is included once per eligible result with its saved fit and parameter summary, while saved electrostatics, protonation, and Spolar outputs are presented without recomputation. Report creation never reruns fitting or advanced analysis.

Write and preview an interpretation

The builder opens in **Interpretation** mode with an editor for your remarks. Text wraps automatically. Interpretation Markdown supports `##` section headings, `###` subsection headings, bullets, `**bold**`, and `*italic*` emphasis. Use the **Interpretation / Preview** control above the workspace to move between writing and the rendered report. Selecting **Preview** automatically builds the PDF when it has not yet been built or is out of date; a current preview is reused. **Update Preview** forces a rebuild. Editing while a preview exists marks it as out of date without repeatedly rendering while you type. **Export PDF...** rebuilds stale content and saves the completed PDF. Results with warnings, outdated inputs, or partially invalid status remain exportable and carry a prominent notice. A structurally unusable result shows a validation error and cannot be previewed or exported.

Generate an automated interpretation

Online generation privacy: Generate sends selected names, comments, conditions, fits and injection evidence, plus your question or context, to the FT-ITC online interpretation service, which uses OpenAI to generate a response. Thermograms start unchecked each time the generator opens and require an explicit opt-in. The service retains usage metadata without automatic expiry; full scientific evidence and generated prose are not included in that metadata. Provider and infrastructure retention are separate; no deletion deadline is guaranteed. Canceling does not retract sent data. Local processing, fitting, saving and export work without generation. Data flow and retention.

The inspector's **Interpretation** section reports whether the text is manual, automatically generated, user edited, or out of date. **Edit** returns focus to the large editor. **Generate interpretation...** opens a dialog with optional **Main question** and **Additional context** fields. Use the question to focus the assessment and the context to explain the experiment. The cross-platform dialog labels its preset selector **Interpretation depth**. Requesting generation sends the selected report evidence and these inputs to the online interpretation service. Compressed thermograms are unchecked each time the generator opens; if **Include compressed thermograms** is available for your access, you can opt in for that request.

Generation assesses all selected results and supporting experiments together, using the same references as the PDF. Results remain independent, including alternative fits of the same experiment. Supporting experiments contribute available observations and metadata without being assigned a fitted result or assumed control role. Generation uses existing processing, integration, residual, and uncertainty evidence where available; it does not rerun integration, baseline fitting, or model fitting. When compressed thermograms are included, they contain minimum and maximum power values for each 15-second interval, plus an aligned fitted baseline when available. Exact peak timing, the order of values within each interval, and the detailed waveform are unavailable. The dialog shows the compact package size against your current

request limit and blocks generation when the estimated request is too large. An opted-in thermogram is retained in the request; a size or model-context error is shown instead of silently omitting it.

When generation succeeds, the desktop app sends an **Interpretation ready** system notification. Whether it appears depends on operating-system notification support and permissions. The draft remains available in the generation dialog for review.

For fitted result members, the interpretation evidence includes the same model-specific c-values shown in **Fit details**. They remain attached to that result member rather than to shared experiment-source evidence. Supporting experiments have no c-value because the report does not assign them a fit.

When several result members have identical source data and processing, the request can share that evidence between them. Each result retains its own fitted values, uncertainty, validity, and constraints. Sharing source evidence does not imply independent replication or equivalent fits.

If the selection contains outdated or failed analyses, a warning gives you an opportunity to review them before continuing. The report distinguishes saved fit inputs and estimates from current observations; some comparisons are unavailable when the saved fit cannot be matched to the current data. An interpretation can identify potential concerns, but it does not certify the experiment or establish a mechanism. The service may suggest references from details such as a paper title, journal, or year, but it does not search the web.

Review and edit the returned draft, then choose **Use in report** to replace the approved interpretation. The draft focuses on observed issues and targeted follow-up checks, using references such as **Result 2** and **Experiment 1B**. It avoids routine warnings based solely on modest deviations in fitted N-values or missing blank titrations.

The interpretation service is asked to keep straightforward assessments to about 500–600 words or fewer, with roughly 1,000 words available for complex collections. These are writing targets, not acceptance limits. Longer text flows onto additional pages without automatic shortening or regeneration. Text length, headings, and Markdown formatting do not prevent you from accepting a draft.

Markdown tables require a header and separator row. They support bold or italic cell text and left, center, or right alignment. Cells wrap, and table headers repeat across pages. Use a few columns to keep tables readable. Unsupported formatting may appear as literal text in the report.

Canceling, closing the dialog, or encountering a service error leaves previously approved text unchanged. Study context, the thermogram setting, approved text, and generation details are saved with the report. Changes to selected results, supporting evidence, context, or settings can mark an approved automated interpretation as out of date. Older approved interpretations remain readable even when their freshness cannot be verified. The report distinguishes manually written interpretations from automatically generated text and subsequent edits.

The **Summary** generation task produces a compact factual report without knowledge-base retrieval or capability-code quota usage. Local report editing, saved interpretations, and PDF export remain available if automated interpretation generation is unavailable.

Save a local copy of interpretation evidence

Use **Save package** in the generation dialog to save a local ZIP of the evidence, question, context, and settings selected for that request. Saving the ZIP does not contact the interpretation service or run new fits. The archive contains scientific data and context; review it before sharing.

Troubleshoot automated interpretation

If generation fails, note the visible error and use **Copy Support Report** when contacting support. Local report editing and PDF export remain available if online generation is unavailable.

Choose the appropriate output tool

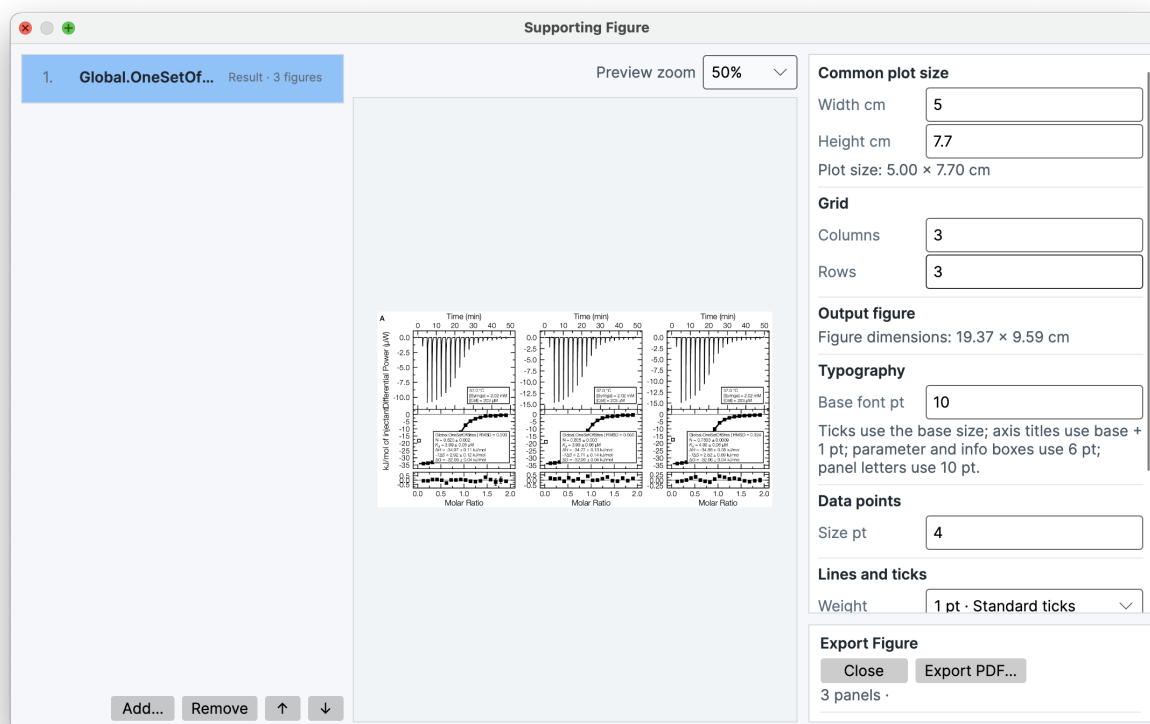
The Analysis Report is distinct from the other output tools:

- **Analysis Result Exporter** writes compact CSV or TSV tables for numerical reuse.
- **Final Figure** writes a publication figure for an experiment, while **Export Associated Final Figures...** writes one figure per result member.
- **Supporting Figure** composes selected figures into one multi-panel canvas.
- **Analysis Report** assembles one or more saved results as self-contained chapters with figures, detailed tables, diagnostics, optional advanced plots, and source and analysis history.

Supporting Figure

Supporting Figure... composes figures from Experiment Data and Analysis Results into a multi-panel canvas. The **Figure order** list defines source order; **Add...**, **Remove**, **Up**, and **Down** change the composition. The source picker can filter available experiment and result figures by name or type.

Common plot size specifies width and height in centimeters. Grid controls specify columns and rows. Typography controls define base font size, point size, line weight, and tick style. **Panel letters** adds stable A, B, ... identifiers. **Panel titles** optionally adds the experiment name after that identifier (for example, **A. Experiment name**) and is off by default for existing Supporting Figure workflows. **Group result figures** and the parameter and information box control affect result grouping and annotations. The preview zoom shows the rendered canvas at 25%, 50%, 75%, or 100%. **Export PDF...** writes the composed supporting figure as a PDF.



Supporting Figure arranges the three member figures from an Analysis Result on a shared multi-panel canvas.

Printing

File > Print prints the active graph or figure through the operating system's print workflow. The active target can be the overview thermogram, processing graph, analysis graph, result graph—including an available **Correlation** matrix—or Final Figure, depending on the selected workspace. The operating-system print dialog supplies the available printer and PDF destinations; the graph content comes from the active application view.

CHAPTER 10

Tools

Design simulated titrations, subtract buffer controls, and merge standard or back-mixed tandem experiments.

Last verified: 2026-09-23

Use **Experiment Designer...** to explore a simulated experiment before collecting data, **Buffer Subtraction...** to account for heat seen in a reference experiment, and **Experiment Merger...** to join consecutive parts of a titration. The sections below explain what each tool takes as input and what it changes in your project.

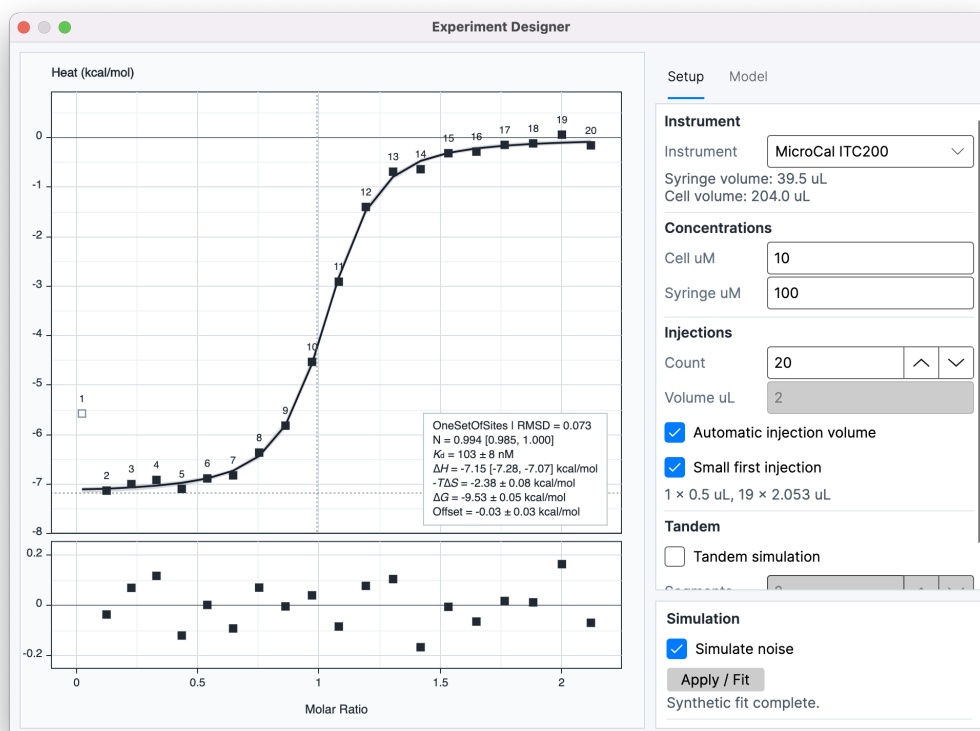
The **Tools** menu contains **Experiment Designer...**, **Buffer Subtraction...**, and **Experiment Merger...**. Each tool handles its output differently: Experiment Designer keeps simulation and fitting inside its window, Buffer Subtraction stores a correction on target experiments, and Experiment Merger creates a new processed Experiment Data item. Source and target selection uses the project state described in Workspace.

Experiment Designer

Experiment Designer... creates a synthetic titration from instrument, concentration, injection, and model parameters. The graph updates as the design changes. The designer does not add the synthetic experiment or its fit as a project item.

Setup and model controls

The **Setup** tab contains **Instrument**, **Cell uM**, **Syringe uM**, injection **Count**, and **Volume uL**. Count and volume define the injection schedule. **Automatic injection volume** derives the injection volume from the selected instrument's standard syringe volume and injection count. **Small first injection** gives the first injection of each load a smaller volume and marks it excluded. The **Simulation** section contains noise controls. **Tandem simulation** creates consecutive loads, with **Segments** setting the number of loads. The designer uses its tandem back-mixing model between loads.



Setup defines the instrument, concentrations, injection schedule, tandem behavior, and simulation state.

The **Model** tab contains **Type**, exposed model **Parameters**, and model-specific **Options**. In the Setup tab, enable **Simulate noise** to add synthetic measurement noise. Its **Noise level** slider ranges from **0.1x** to **5.0x**; **1.0x** is the standard designer noise level. **Apply / Fit** fits the synthetic data in the designer window and reports the fit on its graph; neither the simulation nor this fit becomes an Analysis Result or Experiment Data entry.

Setup **Model**

Model

Type: One-Set-Of-Sites

Parameters

N-value
 N
 0.994042 unitless

Enthalpy
 ΔH
 -715288 kcal/mol

Affinity
 K_d
 0.102797 μM

Simulation

☒ Simulate noise

Apply / Fit

Synthetic fit complete.

Model exposes the synthetic model type and its parameter values.

Buffer Subtraction

BEFORE YOU BEGIN:

Buffer subtraction operates on integrated injection heats. Use experiments with available heats to inspect the correction in the preview; thermogram inputs need processing to supply those heats.

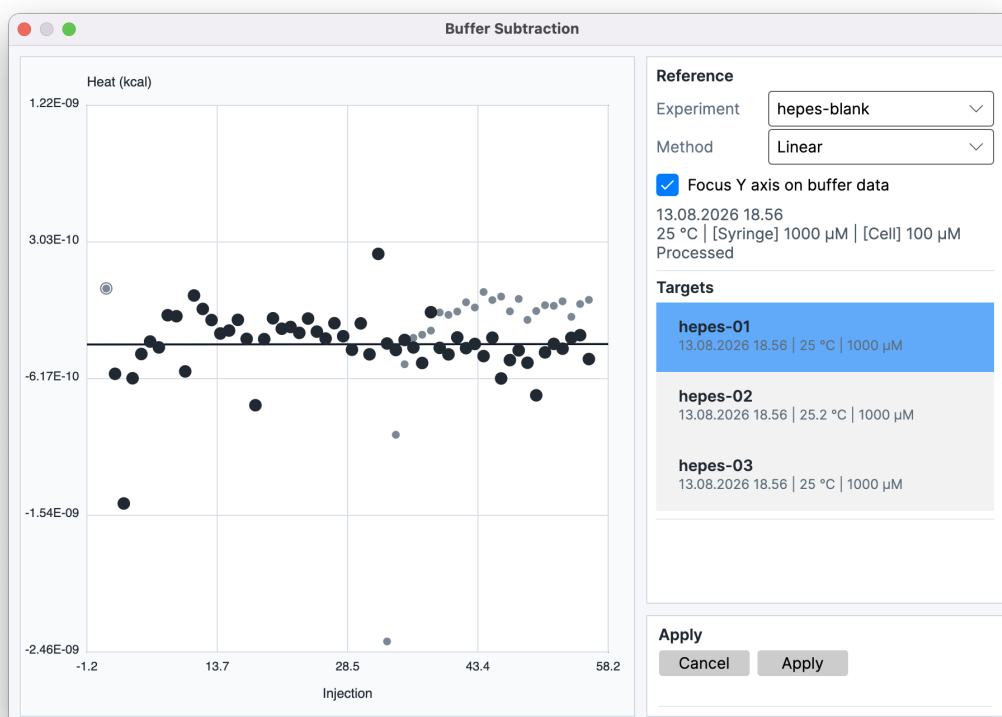
Buffer Subtraction... models background heat from one reference experiment and applies the resulting correction to one or more target experiments. The reference selector shows experiment metadata and a **Processed** or **Not yet processed** status. The target list excludes the selected reference and supports multiple targets. The correction uses the reference's available integrated heats; if the reference is processed later, target corrections update. Processing is described in Processing.

The **Method** selector contains **Matched**, **Linear**, and **Exp. decay**:

- **Matched** evaluates the reference heat at each target injection number, using nearby included reference injections when the matching injection is unavailable.

- **Linear** fits a line through valid, included reference injections and evaluates it across target injections.
- **Exp. decay** fits an exponential-decay model through valid, included reference injections when enough points are available.

The preview graph shows reference and target heats and the selected subtraction model. Reference-point inclusion changes the points available to the fitted methods. **Focus Y axis on buffer data** changes only the preview range. A continuous model line appears for **Linear** and **Exp. decay**; **Matched** is represented by injection-level reference values.



Buffer Subtraction previews the selected reference model and target experiments before storing the correction.

CALCULATION:

$$q_{i, \text{corr}} = q_{i, \text{target}} - q_{i, \text{ref}}$$

$$\sigma_{i, \text{corr}} = \sqrt{\sigma_{i, \text{target}}^2 + \sigma_{i, \text{ref}}^2}$$

The selected method determines the reference value evaluated for injection i . The σ values are standard deviations (SDs). The corrected heat's SD combines the target and reference SDs as independent errors; subtracting the heats does not subtract their SDs.

With **Matched**, the reference SD comes from the matched injection. If the method uses the average of the nearest included injections on either side, its SD is half the square root of the sum of their squared SDs. With **Linear** or **Exp. decay**, the reference SD is estimated from the scatter of included reference heats around the fitted line or curve, rather than from their individual error bars. If the fit has no degrees of freedom left to estimate that scatter, the model assigns a reference SD of zero; this does not establish that the correction is exact.

Apply stores the reference and method on each target. The corrected heats are then used by downstream fitting and export while the original integrated heats remain unchanged. The reference Experiment Data becomes inactive. Changes in its processing or injection inclusion update the target corrections. The subtraction is project data and can affect the validity of dependent results.

Experiment Merger

BEFORE YOU BEGIN:

Select at least two source experiments with thermograms. The merger uses a source's baseline-corrected trace when available and its raw trace otherwise. It processes the newly merged Experiment Data automatically.

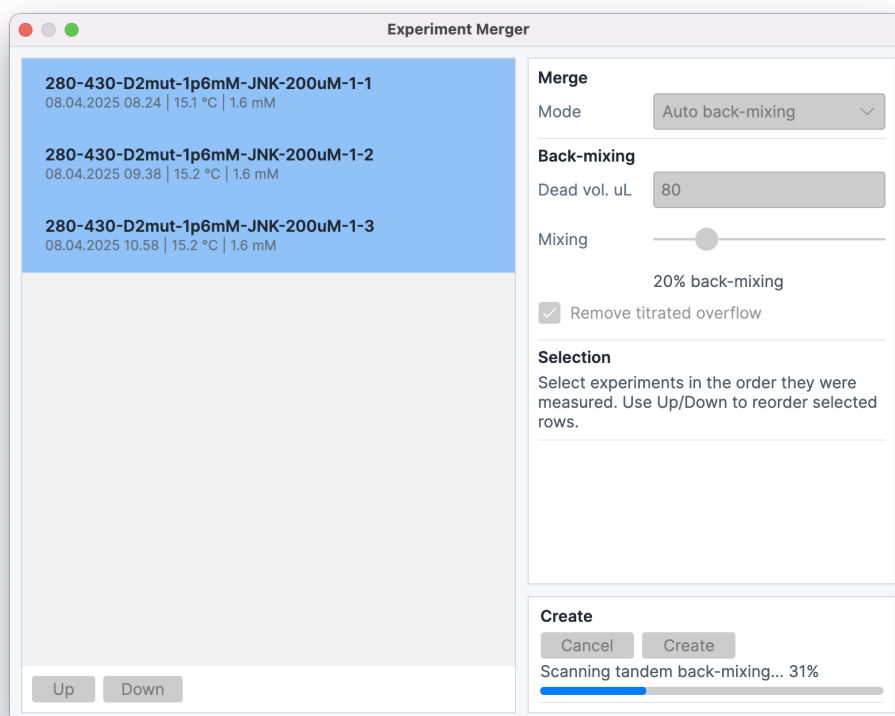
Experiment Merger... joins two or more eligible thermogram experiments from consecutive segments of a tandem titration. The source list contains thermograms that are not already tandem experiments. Selection order defines segment order; **Up** and **Down** reorder selected rows.

The merge **Mode** selector contains:

- **Simple tandem**, which concatenates the segments using the standard concentration progression without a user-selected back-mixing correction.

- **Fixed back-mixing**, which applies one configured mixing fraction at every segment transition. With three or four selected experiments, you can instead set an individual fraction for each reload.
- **Auto back-mixing**, which estimates the reload mixing fractions from the selected experiments and is available for up to five source experiments.

Back-mixing controls include **Dead vol. uL**, the **Mixing** fraction, and **Remove titrated overflow**. Dead volume represents the filling-stem or overflow volume above the active cell volume. The overflow control records whether titrated overflow was removed between segments. In Fixed mode, the shared slider supplies the fraction. With three or four experiments selected, enable the individual-fractions option to set **Reload 1**, **Reload 2**, and, for four experiments, **Reload 3** separately. Auto mode estimates the values from the selected data; these are model-based estimates rather than direct measurements of mixing.



Auto back-mixing scans possible transition corrections for an ordered tandem experiment set.

What happens at a reload

In **Fixed back-mixing** and **Auto back-mixing**, the model tracks the active cell and the dead/overflow volume separately. It starts with the same macromolecule concentration in both and no syringe ligand in either. During a segment, injections change the active-cell concentrations and displaced material enters the dead volume. At the transition to the next segment, the model first removes overflow if **Remove titrated overflow** is selected, then mixes the chosen fraction of the remaining dead volume with the active cell. The resulting active-cell concentrations become the starting concentrations for the next segment. This changes the calculated concentrations used for fitting; it does not add a heat peak or alter the measured thermogram.

CALCULATION:

For each species (macromolecule and syringe ligand), let V_{cell} and C_{cell} be the active-cell volume and concentration just before mixing, V_{dead} and C_{dead} the remaining dead-compartment volume and concentration, and f the selected mixing fraction.

$$V_{\text{mix}} = f \cdot V_{\text{dead}}$$

$$C_{\text{next}} = \frac{V_{\text{cell}}C_{\text{cell}} + V_{\text{mix}}C_{\text{dead}}}{V_{\text{cell}} + V_{\text{mix}}}$$

If overflow is removed, the model first removes a volume equal to the injections delivered in the completed segment, capped at the available dead volume. It assumes that compartment is well mixed, so removal takes the same proportion of each species. The formula then uses the volume and concentrations left behind. A mixing fraction of 0 leaves the active-cell concentrations unchanged; a fraction of 1 includes all remaining dead-compartment liquid in the mixing calculation. The active-cell volume itself stays fixed, and the model retains the remaining material in the dead compartment for later transitions.

Create produces a new processed Experiment Data item. Its thermogram samples are time-shifted and concatenated, its injection sequence retains segment boundaries, and its segment metadata stores the calculated starting active-cell and active-titrant concentrations. The merged item's comments record the selected tandem mode and back-mixing parameters. Source experiments remain separate and are not changed by creation; the new item is marked as a tandem experiment and is not eligible as a later merger source. It is a snapshot and does not update if its source experiments are subsequently edited. The resulting item can be fitted through Analyze Data.

The new experiment uses the **Injection bookkeeping** method selected in **Preferences > Processing** for dilution during each injection. That choice is separate from the merger's between-segment back-mixing setting. See Injection bookkeeping for the methods and their limits.

INTERPRETATION:

A configured or automatically fitted back-mixing fraction is a model-based correction, not a direct measurement of liquid mixing that occurred between runs.

CHAPTER 11

Settings and defaults

Configure shared defaults for display, processing, fitting, and export.

Last verified: 2026-09-23

Preferences set the starting behavior for future work, such as display units, processing choices, and export formatting. Changing a default does not by itself redo an existing fit or processing decision saved in a project.

Preferences... contains **General**, **Processing**, **Fitting**, and **Export**. Settings labeled as defaults provide starting values for new or reset work. Display settings affect presentation; export settings affect generated tables and figures. Project-specific values stored in a project remain distinct from application preferences. Project and recovery behavior is covered in Installation, files, and projects.

Both desktop applications use the same built-in values for first launch and Restore Defaults, including a **20,000-iteration optimizer limit** and **10% automatic concentration SD**. Saved custom values remain in effect, including values between or beyond the slider presets. Choosing **Apply** also preserves settings that are not shown in that application's Preferences window.

Restore Defaults stages the built-in values in the window. **Apply** saves the staged values as application preferences. **Cancel** closes the window without saving staged edits.

General

SETTING	EFFECT
Energy units	Select Joule or Calories . Displayed values automatically use the base or kilo prefix from the finite central values being shown; empty groups default to kJ or kcal.
Concentration unit	Shared default unit for concentration entry, parameter display, and supported result-table concentration fields; format-specific data exports use their documented units.
Designer instrument	Shared default instrument for the Experiment Designer and its instrument-specific volumes.
Number precision	Controls numeric presentation: Strict , Standard , Single decimal , or All decimals . It does not set export decimal places.

Uncertainty display	Shared uncertainty presentation: Automatic , Standard deviation , Confidence interval , SD + confidence interval , or None . Automatic uses the confidence interval for a quantity whose stored 95% interval is materially asymmetric around its best-fit value; otherwise it uses SD. This changes presentation, not the underlying fit or calculated uncertainty.
Reference temperature (°C)	Temperature used where a result or derived quantity is evaluated at a reference temperature.
Minimum temperature span (°C)	Minimum temperature variation required for temperature-dependent result analyses.
Minimum salt span (mM)	Minimum ionic-strength variation required for salt-dependent result analyses.
Include buffer in ionic-strength calculation	Includes buffer contribution when ionic strength is calculated.
Check for updates and online resources on launch	Controls launch-time online checks; it does not disable local analysis.
Confirm remove/delete actions	Controls confirmation before remove or delete commands.
Automatically discard injections outside the thermogram range	Controls automatic removal of orphan injections when data are loaded.
Enable autosave	Enables periodic recovery copies.
Interval (minutes)	Sets the autosave interval. The built-in default is 5 minutes .
Maximum files	Sets the number of autosave files retained. The built-in default is 10 files .

Prompt to recover after an interrupted session	Controls whether the application presents an available autosave recovery after an interrupted session. Recovery prompting is enabled by default.
Open Autosave Folder	Opens the folder containing autosave recovery files.

Energy family, concentration unit, number precision, and uncertainty presentation are shared application settings where the corresponding control is available. The energy family changes displayed values but does not change the units used by supported interchange formats. Automatic display uses kJ or kcal when a value group is empty, zero, non-finite, or at or above the 100-unit threshold; fixed publication/result-export overrides are configured in their respective dialogs.

Processing

SETTING	EFFECT
Injection bookkeeping	Selects MicroCal (default), Ideal continuous mixing or Discrete displacement for new data. Ideal continuous mixing models exponential concentrations and accounts for displaced heat along the mixing trajectory; Discrete displacement uses discrete replacement without numerical integration. Change an existing ordinary experiment explicitly in Experiment Details; rebuild a tandem experiment to change its method. Saved fits are not automatically migrated.
Buffer subtraction	Sets the default buffer-subtraction model: Matched , Linear , or Exp. decay .
Discard integration regions for baseline	Controls whether existing integration regions are excluded from baseline construction.
Reprocess integrated heats on load	Sets the initial Recalculate concentrations and ratios choice for <code>.dat</code> and <code>.aff</code> imports. You can change it for an import without changing this preference. Recalculation uses the imported injection volumes and experiment concentrations; it does not create a thermogram or repeat baseline correction and peak integration. <code>.dh</code> concentrations are always calculated.
Point density	Sets the default spline point density: Sparse , Balanced , or Dense .
Handle mode	Sets the default spline handle calculation: Mean or Median in the preferences window.
Allow spline point time dragging by default	Controls whether spline points can be moved in time by default.
Copy integration start with selected region	Controls whether copying an integration region also copies its start time.

Processing preferences provide defaults for new processors. Processing values already stored with an experiment are not replaced merely by changing a preference.

Fitting

SETTING	EFFECT
Default solver	Sets the starting optimizer: Nelder-Mead [SIMPLEX] or Levenberg-Marquardt .
Error estimation	Sets the default uncertainty method: None , Bootstrap residuals , Leave-one-out , or Profile likelihood . The built-in method is Bootstrap residuals .
Bootstrap iterations	Sets the number of residual-bootstrap refits. Leave-one-out uses one refit per deletion and profile likelihood does not use this count. The built-in count is 100 .
Optimizer tolerance	Sets the solver tolerance preset: Fast , Relaxed , Balanced , Strict , or Very Strict . The built-in default is Balanced .
Max iterations	Sets the maximum number of optimizer iterations. The built-in default is 20,000 , both at first launch and after Restore Defaults is applied.
Parameter limits	Sets the default parameter-limit policy: Standard , Extended , or No limit .
Use injection-error weighted fitting	Controls weighting of injection observations by their estimated errors.
Include concentration uncertainty in bootstrap	Includes concentration uncertainty in residual-bootstrap resampling. Leave-one-out keeps concentrations fixed.
Automatic concentration SD (%)	Sets the automatic fractional concentration SD used when concentration-uncertainty handling is enabled.
Create single-experiment analysis result	Controls creation of an Analysis Result after a usable single-experiment fit. Disabled in the built-in defaults.
Create global analysis result	Controls creation of a combined Analysis Result after a usable multiple-experiment fit. Enabled in the built-in defaults.
Auto-open new analysis result	Controls whether a newly created result is opened automatically.

Bootstrap method and count are shared fitting defaults; concentration sampling and parameter unlocking apply only to residual bootstrap. If profiling is canceled or fails, the successful primary fit is retained. Whether an update replaces an existing result with partial profile output depends on the stored update policy. Fit-specific settings captured in an Analysis Result remain part of that result. See Single-experiment fitting for model and uncertainty interpretation.

Automated interpretation access

To request an account with additional interpretation options, use the FT-ITC interpretation service registration page. If eligible, submit your name and email, accept the Terms and acknowledge the Privacy Notice, then verify your email; the access code is sent separately by email. Keep the code confidential. If it is exposed or does not arrive, contact support@ft-itc.org.

Without a code, automated interpretation uses the public **Default** depth. Enter the account code and choose **Verify Access** to load the options available to your account. When additional presets are available, choose one from **Interpretation depth**. Preferences show available account details and remaining quota. The code stays masked and is checked when an interpretation is generated.

Export

SETTING	EFFECT
Selection	Sets the default export scope: Selected experiment , Active experiments , or All experiments .
Decimals	Sets the number of decimal places in exported numeric tables.
Export baseline-corrected data	Includes baseline-corrected data in data exports.
Export fit points with peaks	Includes fitted peak points with exported peak data.
Molar ratio	Includes molar-ratio values in exported tables.
Injection info	Includes injection volume, delay, peak error, integration length, and temperature columns.
Concentrations	Includes cell and syringe concentration columns.
Included state	Includes the injection-inclusion state.
Peak heats	Includes integrated peak heats.

Fit values	Includes fitted values.
Width cm and Height cm	Set the default final-figure dimensions. The built-in dimensions are 6.5 × 10 cm .
Publication font	Selects the publication figure font where the platform provides this selector.
Show residual graph	Includes the residual graph in final figures.
Show residual graph gap	Includes the gap separating residual and fit panels.
Unify residual graph axis	Uses a common residual-axis scale across applicable panels.
Fit line	Sets fit-line smoothing: Smooth , Spline , or Linear .
Show parameter box by default	Includes the parameter information box in new final figures.
Show experiment details by default	Includes experiment metadata in the figure information box.
Show model info by default	Includes model information in the figure information box.
Auto axes ignore excluded/bad points	Excludes points marked excluded or bad when automatic axes are calculated.
Thermodynamic parameters	Includes thermodynamic parameters in final-figure information.
Offset parameter	Includes the fit offset in final-figure information.
Derived parameters	Includes derived parameters in final-figure information.
Temperature	Includes temperature in final-figure information.
Concentrations	Includes concentrations in final-figure information.
Injection delay	Includes injection delay in final-figure information.
Instrument	Includes instrument information in final-figure information.
Attributes	Includes experiment attributes in final-figure information and limits

their display to **Used in analysis**, **All**, or **None**.

Export preferences affect newly generated exports and figure defaults; they do not rewrite an existing export or a stored figure configuration. See Figures and export for output formats and units.

The energy-unit preference also sets the initial molar-energy unit in the Integrated Peaks, Combined Data, and ITCsim export dialogs: kJ/mol for Joules or kcal/mol for Calories. You can choose another unit for each export without changing the preference.

PLATFORM NOTE:

The **Publication font** selector is available on Windows and Linux, with choices of **Native**, **Inter**, and **Liberation Sans**. Some macOS releases use the native publication renderer and do not show this selector.

CHAPTER 12

Glossary and equations

Terminology, keyboard conventions, calculation symbols, equations, and references.

Last verified: 2026-09-23

Use this page to look up a term or symbol while reading the task chapters. The equations specify what the application calculates; you do not need them to follow the Quick start.

Workspace terms

TERM	MEANING IN FT-ITC ANALYSIS
Experiment Data	One imported or application-created ITC dataset, including its details and analysis state.
Data / Results	The workspace list containing experiments and stored Analysis Results.
Selected	The item currently shown or targeted by a selection-specific command.
Active experiments	Experiments included in a multi-experiment operation.
Solution	A fitted model state for one experiment.
Analysis Result	A stored fit result with one or more member solutions and the fit state used to create it.
Attribute	Experiment metadata used for organization or analysis context.
Valid result	An Analysis Result whose recorded fit inputs still match the current project state.
Correlation view	Pearson correlations between fitted parameter coordinates across residual-bootstrap refits that have finite values for every displayed parameter, with diagnostics for simulation precision.
Pearson correlation	A value from -1 to $+1$ describing linear co-variation between two fitted coordinates.
Monte Carlo precision interval	Approximate Fisher-transformation interval describing how precisely the correlation coefficient r is estimated from the finite retained-refit ensemble; it is not uncertainty for either fitted parameter or a model-validity assessment.
AIC / AICc	Analysis-level criteria for comparing saved fits with the same observations, response definition, weighting mode, and likelihood convention; smaller values are preferred. Both modes estimate one residual-variance parameter; weighted criteria use integration errors as relative uncertainties. AICc uses the standard small-sample approximation for nonlinear fits and is preferred when available. Independently fitted members also show a per-member AICc / AIC column for same-experiment comparisons. Neither criterion establishes model adequacy.

Uncertainty terms

TERM	MEANING IN FT-ITC ANALYSIS
Best-fit value	Parameter value from the primary optimization. Resampling estimates uncertainty around this value but does not replace it with a resampling mean or median.
Standard deviation (SD)	Root-mean-square deviation of retained resampling values from the primary best-fit value.
95% confidence interval (CI)	The 2.5th and 97.5th percentiles of the retained resampling distribution. It can be asymmetric around the best-fit value.
Automatic uncertainty display	Shows CI for a materially asymmetric stored interval and SD otherwise. It selects a presentation separately for each reported quantity.
Displayed parameter uncertainty	Fitted-coordinate uncertainty after conversion or propagation into the quantity shown to the user. The Automatic display decision is applied to this displayed quantity.
Profile likelihood	Conditional profiling at a fixed 95% confidence level. Unweighted and weighted intervals use an F-calibrated worsening relative to the best residual score. Weighted intervals retain processing-derived peak-area SDs as unchanged relative injection weights and estimate the overall residual scale separately. A parameter bound reached before crossing is censoring, not a confidence endpoint. The reported primary fit remains unchanged.
Profile equivalent SD	Symmetric <code>value ± SD</code> display scale computed from a complete asymmetric profile interval; it is a display equivalent, not a Gaussian sampling SD.

Keyboard conventions

The application command modifier is **Command** on macOS and **Ctrl** on Windows or Linux.

SHORTCUT	FUNCTION
Command/Ctrl + O	Open data or a project.
Command/Ctrl + S	Save the current project.
Command/Ctrl + Shift + S	Save the project under another name.
Command/Ctrl + E	Open data export.
Command/Ctrl + P	Print the active graph or figure.
Command/Ctrl + Z	Restore the most recently removed item.
Command/Ctrl + I	Invert the Active Experiment Data set.
Command/Ctrl + comma	Open Preferences.
Command/Ctrl + Q	Quit FT-ITC Analysis.
F1	Open Help and Guide on Windows and Linux. On macOS, open Help through the ? menu command.
Enter	Open Details for the selected Data/Results item on Windows and Linux.
Left / Right	Change the selected injection in Process Data .
Space	Copy the selected integration length to the next injection in Process Data ; the start is also copied when that processing option is enabled.

Thermodynamic symbols

SYMBOL	MEANING
N	Stoichiometry or site-number term under the selected model.
α	Syringe-side active-concentration correction when syringe correction is enabled.
K_a	Association constant.
K_d	Dissociation constant.
K_i	Macroscopic association constant for sequential transition i .
β_i	Cumulative sequential association product $\prod_{j=1\dots i} K_j$.
F_i	Fraction of macromolecule in sequential state MX_i .
$\bar{v}$	Mean ligand occupancy in the sequential model.
ΔH	Binding enthalpy change.
ΔG	Gibbs free-energy change.
$-T\Delta S$	Entropic contribution reported by the application at the evaluation temperature.
ΔC_p	Heat-capacity change in a supported temperature-dependent analysis.
q	Integrated injection heat.
σ	Standard uncertainty or spread, according to context.
Offset	Energy-per-mole-of-injectant correction included in the modeled injection heat.

Calculation symbols

SYMBOL	MEANING
$P(t)$	Differential power at time t in a thermogram.
$b(t)$	Estimated baseline at time t .
n_i	Amount injected in injection i .
r_i	Residual for injection i : observed heat minus model heat.
r_{jk}	Pearson correlation between fitted parameter coordinates j and k .
θ_{bj}	Value of fitted coordinate j in complete bootstrap refit b .
σ_i	Processing-derived uncertainty for injection i .
R	Gas constant.
T	Absolute temperature, in kelvin, for thermodynamic relationships.
I	Ionic strength used in salt analysis.
a_{ion}	Ion activity used in Counter Ion Release analysis.
m	Fitted slope in the Protonation relationship; the application reports Protons as $-m$.
n_{ion}	Counter-ion slope reported by Counter Ion Release analysis.
RSS	Raw residual sum of squares, $\sum r_i^2$, over included injections.
Q	Standardized residual sum of squares, $\sum (r_i/\sigma_i)^2$. Weighted AIC/AICc estimates a common variance multiplier as Q/n .
L	Gaussian likelihood used for AIC/AICc.
p	Number of free fitted parameters in the saved global model; shared coordinates count once and member coordinates count per member.
K	AIC/AICc likelihood parameter count: $p + 1$ in both weighting modes, including one estimated common residual variance or one estimated multiplier of the injection-error variances.

Equation index

RELATIONSHIP	LOCAL EXPLANATION
Integration and molar normalization	Processing
Dilution-driven dissociation of an associated pair	Dissociation
Sequential binding polynomial and ligand balance	Sequential Binding Sites
Thermodynamic conversion and -TΔS	Thermodynamic relationships
Fit residuals, RMSD, and weighting	Weight by injection error
AIC, AICc, Gaussian likelihood, and parameter counting	Information criteria
Temperature-dependent constraints	Multiple-experiment fitting
Parameter correlation (Pearson residual bootstrap)	Parameter correlation
Bootstrap SD and percentile confidence interval	Parameter uncertainty
Automatic uncertainty display	Uncertainty and evaluation temperature
Salt dependence	Salt
Protonation dependence	Protonation
Buffer correction	Buffer Subtraction
Tandem reload mixing	What happens at a reload

Units are those shown by the application or export.

References

- FT-ITC Analysis repository
- FT-ITC Analysis website
- FT-ITC Project Viewer
- Software DOI: 10.5281/zenodo.14832177
- Malvern Instruments, *MicroCal ITC Analysis Software Using Origin User Manual*, MAN0577-02-EN-00 (20 May 2015), section 12.3.1, equations 2 and 4.
- **Help > Citation** for the current paper citation, versioned software citation, and BibTeX
- Philippe Dumas, *Isothermal titration calorimetry in the single-injection mode with imperfect mixing*, European Biophysics Journal 51, 77–84 (2022). The application's **Ideal continuous mixing** option is inspired by the paper but uses ideal mixing with a finite-injection numerical approximation; it does not implement the paper's imperfect-mixing model.
- Ernesto Freire, Arne Schön and Adrián Velazquez-Campoy, *Isothermal titration calorimetry: general formalism using binding polynomials*, Methods in Enzymology 455, 127–155 (2009). The application's **Discrete displacement** option uses this convention for concentrations and displaced-volume heat compensation, retaining FT-ITC's equilibrium solvers and offset convention.

Scientific claims and model-specific methods require the primary literature appropriate to the experiment and analysis.